Informatics Institute of Technology

In Collaboration with

Robert Gordon University, Aberdeen, Scotland

AI-Driven Dynamic Narrative Personalization in Games:

A Closed-Loop Plugin Using Hexad Player Modeling, RAG-Grounded LLM Generation, and PPO Reinforcement Learning

Final Year Thesis by

Rivindu Bopage

RGU ID – 2312553

IIT ID – 20220580

Supervised by

Bimali Wickramasinghe

April 2026

Submitted in partial fulfilment of the requirements for the BSc. (Hons) in Artificial Intelligence and Data Science at the Robert Gordon University, Aberdeen, UK.

DECLARATION

I confirm that the work contained in this BSc project report has been composed solely by myself and has not been accepted in any previous application for a degree. All sources of information have been specifically acknowledged and all verbatim extracts are distinguished by quotation marks.

Date: 2026-04-04

Student's Name: Rivindu Bopage

The above student carried out his research project under my supervision.

Date: 2026-04-05

Supervisor's Name: Bimali Wickramasinghe

ROBERT GORDON UNIVERSITY•ABERDEEN

STUDENT PROJECT ETHICAL REVIEW (SPER) FORM

The aim of the University's *Research Ethics Policy* is to establish and promote good ethical practice in the conduct of academic research. The questionnaire is intended to enable researchers to undertake an initial self-assessment of ethical issues in their research. Ethical conduct is not primarily a matter of following fixed rules; it depends on researchers developing a considered, flexible and thoughtful practice.

The questionnaire aims to engage researchers discursively with the ethical dimensions of their work and potential ethical issues, and the main focus of any subsequent review is not to 'approve' or 'disapprove' of a project but to make sure that this process has taken place.

The *Research Ethics Policy* is available at
www.intranet.rgu.ac.uk/credo/staff/page.cfm?pge=7060

Student Name	Rivindu Bopage
Supervisor	Ms. Bimali Wickramasinghe
Project Title	Dynamic, Personalized Quest and Dialogue Generation for Open-World Games Using AI and Player Modeling
Course of Study	BSc in AI and Data Science
School/Department	School of computing

Part 1 : Descriptive Questions			
1	Does the research involve, or does information in the research relate to:	Yes	No
	(a) individual human subjects		✔
	(b) groups (e.g. families, communities, crowds)		✔
	(c) organisations		✔
	(d) animals?		✔
	Please provide further details:		
2	Will the research deal with information which is private or confidential?	Yes	No
			✔
	Please provide further details:		

Part 2: The Impact of the Research			
3	In the process of doing the research, is there any potential for harm to be done to, or costs to be imposed on	Yes	No
	(a) research participants?		✔
	(b) research subjects?		✔
	(c) you, as the researcher?		✔
	(d) third parties?		✔
	Please state what you believe are the implications of the research:		
4	When the research is complete, could negative consequences follow:	Yes	No
	(a) for research subjects		✔
	(b) or elsewhere?		✔
	Please state what you believe are the consequences of the research:		

Part 3: Ethical Procedures			
5	Does the research require informed consent or approval from:	Yes	No
	(a) research participants?		✔
	(b) research subjects		✔
	(c) external bodies		✔
	If you answered yes to any of the above, please explain your answer:		
6	Are there reasons why research subjects may need safeguards or protection?	Yes	No
			✔
	If you answered yes to the above, please state the reasons and indicate the measures to be		
7	Has PVG membership status been considered?		
	(a) PVG membership is not required.	✔	
	(b) PVG membership is required for working with children.		
	(c) PVG membership is required for working with protected adults.		
	(d) PVG membership is required for working with both children and protected		
	If you answered yes to (b), (c) or (d) above, please give details:		
8	Are specified procedures or safeguards required for recording, management, or storage of data?	Yes	No
			✔
	If you answered yes to the above, please outline the likely undertakings:		

Part 4: The Research Relationship			
9	Does the research require you to give or make undertakings to research participants or subjects about the use of data?	Yes	No
			✔
	If you answered yes to the above, please outline the likely undertakings:		
10	Is the research likely to be affected by the relationship with a sponsor, funder or employer?	Yes	No
			✔
	If you answered yes to the above, please identify how the research may be affected:		

Part 5: Other Issues			
11	Are there any other ethical issues not covered by this form which you believe you should raise?	Yes	No
			✔

Statement by Student I believe that the information I have given in this form is correct, and that I have addressed the ethical issues as fully as possible at this stage.			
Signature		Date	14/11/2025

If any ethical issues arise during the course of the research, students should complete a further Student Project Ethical Review (SPER) form.

The *Research Ethics Policy* is available at www.intranet.rgu.ac.uk/credo/staff/page.cfm?pge=7060

Part 6: To be completed by the supervisor			
12	Does the research have potentially negative implications for the University?	Yes	No ✔
	If you answered yes to the above, please explain your answer:		
13	Are any potential conflicts of interest likely to arise in the course of the research?	Yes	No ✔
	If you answered yes to the above, please identify the potential conflicts:		
14	Are you satisfied that the student has engaged adequately with the ethical implications of the work? [In signifying agreement, supervisors are accepting part of the ethical responsibility for the project]	Yes ✔	No
	If you answered no to the above, please identify the potential issues:		
15	**Appraisal:** Please select one of the following		
	The research project should proceed in its present form – no further action is required	✔	
	The research project requires ethical approval by the School Ethics Review Panel		
	The research project needs to be returned to the student for modification prior to further action		
	The research project requires ethical review by an external body. If this applies please give details		
	Title of External Body providing ethical review		
	Address of External Body		
	Anticipated date when External Body may consider project		

Affirmation by Supervisor			
I have read the student's responses and have discussed ethical issues arising with the student. I can confirm that, to the best of my understanding, the information presented by the student is correct and appropriate to allow an informed judgement on whether further ethical approval is required.			
Signature	Bimali	**Date**	21/11.2025

ABSTRACT

Narrative personalization in games (the capacity of a game's story to adapt meaningfully to individual player behaviour) remains an under-explored intersection of artificial intelligence, human–computer interaction, and game design. Most commercial games deliver identical narrative content regardless of how a player engages with the world, foregoing the well-established link between personalized experience and intrinsic motivation.

This dissertation presents the design, implementation, and evaluation of a closed-loop Unity plugin for real-time dynamic narrative personalization. The system integrates five tightly coupled modules: (1) a telemetry collector that streams 23 behavioural signals from the game client every five seconds; (2) a player modelling service that derives an 11-feature representation, a six-dimensional continuous Hexad player-type profile, and a Flow state classification (FLOW, BOREDOM, ANXIETY, APATHY) from those signals; (3) a narrative generation service that employs Retrieval-Augmented Generation (RAG) over a structured lore corpus using the sentence-transformers all-MiniLM-L6-v2 model together with an episodic session memory store; (4) an adaptation engine based on Proximal Policy Optimization (PPO), trained via Stable-Baselines3 over a custom Gymnasium Markov Decision Process environment; and (5) a Unity Content Injection Module that surfaces the generated narrative inside a 3D RPG testbed built on the AnyRPG Core framework.

A key novelty is the derivation of Hexad player-type scores entirely from gameplay telemetry, without requiring the player to complete a questionnaire, enabling continuous, within-session profiling. The PPO agent selects among eight narrative actions (e.g., PROVIDE_GUIDANCE, ADD_MYSTERY, INCREASE_URGENCY) based on the current Flow state and Hexad profile, with a reward function shaped around challenge–skill balance. A cold-start heuristic governs the first 120 seconds before the learned policy activates.

The system is evaluated through a between-subjects user study ($N = 50$; 25 Experimental, 25 Control). The primary dependent variable is the Flow subscale of the Game Experience Questionnaire (GEQ); secondary measures include the GEQ Immersion subscale, the custom three-item Narrative Personalization Scale (NPS-3), and the miniPXI player experience inventory. Qualitative data are gathered through semi-structured interviews analysed using Reflexive Thematic Analysis (Braun and Clarke, 2022). Results are reported in Chapter 4 and discussed in Chapter 5.

The work makes six original contributions: telemetry-derived Hexad profiling, RAG-grounded narrative generation, episodic session memory in the LLM prompt, PPO for narrative action selection, a Flow-based MDP reward, and a complete closed-loop integration of all five modules within a single game plugin.

Keywords: narrative personalization, player modelling, Hexad typology, Flow theory, reinforcement learning, large language models, retrieval-augmented generation, game experience, Unity plugin.

ACKNOWLEDGEMENTS

I would like to express my gratitude to my supervisor, Bimali Wickramasinghe, who pushed back on vague ideas until they became concrete ones and gave honest feedback when I needed it most. This project would look very different without her guidance.

Thanks are due to the staff at Informatics Institute of Technology and Robert Gordon University, and to the participants who gave up their time to play through the testbed game and answer my questions afterward.

This system stands on the shoulders of several open-source projects (AnyRPG Core, FastAPI, Stable-Baselines3, Gymnasium, Ollama, and sentence-transformers), and I am grateful to the communities that build and maintain them.

Finally, thank you to my family and friends for their unwavering support and patience throughout this journey.

CONTENTS

LIST OF FIGURES

LIST OF TABLES

LIST OF ABBREVIATIONS

AI	Artificial Intelligence
API	Application Programming Interface
CHI	Conference on Human Factors in Computing Systems
CORS	Cross-Origin Resource Sharing
DDA	Dynamic Difficulty Adjustment
DPM	Damage Per Minute
DV	Dependent Variable
EM-LLM	Episodic Memory Large Language Model
GEQ	Game Experience Questionnaire
HTTP	Hypertext Transfer Protocol
JSON	JavaScript Object Notation
LIGS	LLM-Integrated Game Storytelling
LLM	Large Language Model
LTS	Long-Term Support
MDP	Markov Decision Process
miniPXI	Mini Player Experience Inventory
NLP	Natural Language Processing
NPC	Non-Player Character
NPS-3	Narrative Personalization Scale (3-item)
PANGeA	Procedural Adaptive Narrative using a Game Architecture
PPO	Proximal Policy Optimization
RAG	Retrieval-Augmented Generation
REST	Representational State Transfer
RL	Reinforcement Learning
RPG	Role-Playing Game
SDT	Self-Determination Theory
SQL	Structured Query Language
TMPro	TextMeshPro
UI	User Interface
UPM	Unity Package Manager
WS	WebSocket

Chapter 1

INTRODUCTION

1.1 Chapter Overview

This chapter motivates the research, states the problem, and defines the questions and objectives that guide the rest of the thesis. It opens with the problem that current game narrative systems are designed to solve imperfectly, explains why solving it matters, identifies what prior work has left undone, and lays out the research questions, objectives, significance, and scope that frame the investigation.

The remainder of the thesis is organised as follows. Chapter 2 surveys the literature on narrative personalization, player modelling, Flow theory, LLMs for games, RL-based adaptation, and episodic memory. Chapter 3 sets out the methodology, covering research design, system architecture, module design, implementation, testbed game, and evaluation plan. Chapter 4 reports the user study results. Chapter 5 discusses the findings, contributions, limitations, and threats to validity. Chapter 6 concludes with a summary, limitations, future work, and a reflective account of the research process. Questionnaire instruments, the API specification, and the MDP formal specification are collected in the appendices.

1.2 Problem Statement

The central problem this research addresses is straightforward: **current game narrative systems are static: they do not adapt their tone, urgency, or content to the psychological state and demonstrated preferences of an individual player in real time.** The result is a persistent mismatch between what a player needs from the story and what the story actually delivers. A player drifting into boredom, whose challenge–skill ratio has dropped below a meaningful threshold, continues to receive the same measured, low-stakes descriptions that a fully engaged player receives. A player overwhelmed by challenge, tipping into anxiety, gets no additional guidance from the narrative world. A player whose behaviour signals deep curiosity about lore encounters the same sparse environmental text as one who ignores it entirely.

The cost of this mismatch is twofold: reduced engagement in the short term, and a missed opportunity to use narrative as a lever for nudging players back toward optimal experience. Self-Determination Theory (Ryan and Deci, 2000) predicts that perceived competence, autonomy, and relatedness sustain intrinsic motivation, and a story that responds to how someone plays is a direct vehicle for signalling all three.

Solving this problem, however, is not trivial. Reliable real-time inference of a player's psychological state demands robust signal processing over noisy behavioural telemetry. LLM-generated narrative in a game context must be tightly constrained, because a single anachronism, fourth-wall break, or lore contradiction can shatter fictional immersion. And the mapping from player state to the right narrative response is too complex and context-dependent to be fully hand-authored; it must be learned.

1.3 Research Motivation

Games hold a distinctive place among interactive media: they are the only narrative format in which the audience both directs the protagonist and actively shapes how the story unfolds. However, despite the immense expressive possibilities that this interactivity offers, most commercially released games present nearly the same narrative sequence to every player. The village elder always delivers the same warning, the antagonist repeats the same monologue, and the quest log lists the same objectives whether the player is a careful explorer who pores over every scrap of environmental lore or a reckless disruptor who skips dialogue and rushes straight into battle.

This uniformity does not stem from a lack of interest on the part of game designers (narrative branching is widely acknowledged as a difficult craft problem) but from the structural realities imposed by the cost of creating alternatives. Each branch, each tailored line or outcome, demands additional writing, voice work, and implementation, all of which grow at a combinatorial rate. As a result, even the most narratively ambitious games (for example, CD Projekt Red's *Cyberpunk 2077* or Larian Studios' *Baldur's Gate 3*) hit a practical upper limit on authored branching long before they can achieve genuinely player-adaptive personalization.

The rise of large language models (LLMs) that can produce coherent, long-form natural language, together with progress in real-time player behaviour modelling and reinforcement learning-based policy optimisation, points to a new opportunity: narrative that is *generated* on the fly, rather than pre-authored, in response to an individual player's observed preferences, cognitive state, and engagement profile (Brown et al., 2020). If such a system can be made reliable, constrained to lore-consistent content, and sensitive to psychological indicators of engagement, it would fundamentally alter how players encounter game stories, not by offering better-written branches, but by telling stories that actively notice and react to the player.

This work is driven by three key observations. Self-Determination Theory (Ryan and Deci, 2000) indicates that perceived competence, autonomy, and relatedness are foundational to intrinsic engagement and narrative personalisation directly communicates all three. At the same time, the commercial games industry has poured resources into player analytics and telemetry pipelines, yet these data streams are used almost entirely for monetisation and retention tracking, rather than for adapting content in real time. Finally, the convergence of accessible RL libraries (Stable-Baselines3; Raffin et al., 2021), open-weight LLMs (Phi-3.5 Mini; Microsoft, 2024), and modern semantic search methods (sentence-transformers; Wang et al., 2020) makes implementing such a system realistically achievable within an undergraduate project.

This dissertation tackles that opportunity. It introduces a closed-loop plugin, designed for integration with Unity, that continuously profiles the player using a telemetry-driven Hexad player-type model (Tondello et al., 2016) and a Flow state classifier based on Csikszentmihalyi's (1990) challenge–skill balance framework. A Proximal Policy Optimization (PPO; Schulman et al., 2017) agent chooses among eight possible narrative modulation actions in response to these live signals. A Retrieval-Augmented Generation (RAG) pipeline constrains the LLM's output to a structured lore database, while an episodic session memory module inspired by EM-LLM (Fountas et al., 2024) maintains narrative coherence over the course of a play session. The system is then compared with a fixed, non-adaptive narrative baseline in a between-subjects user study, using established psychometric scales to assess Flow, immersion, and perceived personalisation.

1.4 Research Gap

Prior approaches to narrative personalization in games cluster into three broad categories, and each leaves a gap that the present work is designed to fill.

Authored branching systems, such as those found in Mass Effect, The Witcher series, and Disco Elysium, create the impression of personalization through meaningful player choices, but the narrative they deliver is ultimately deterministic with respect to choice history. They do not respond to behavioural signals like play pace, combat avoidance, or exploration rate; they respond only to explicit decisions made within a pre-authored decision space.

Procedural narrative generation systems, including Tracery (Compton et al., 2015) and the Expressionist framework (Kreminski et al., 2019), showed that grammatical expansion could yield contextually varied text. Yet these systems still depended on extensive hand-authored production rules and had no mechanism for sensing or responding to the player's psychological state.

LLM-based narrative generation systems represent the most recent wave. PANGeA (Todd et al., 2024) demonstrated that large language models can produce coherent, in-world narrative when given adequate contextual grounding, but PANGeA includes neither a real-time player model nor an adaptive policy; its generation is procedural (driven by game state) rather than personalized (driven by player psychology). Similarly, Kumaran and Riedl (2025) explored LLM integration into game narrative pipelines with a focus on coherence, but without player modelling or reinforcement learning-based adaptation.

No prior work has integrated continuous Hexad profiling derived from behavioural telemetry, Flow-based RL reward shaping, RAG-grounded LLM generation, and episodic session memory into a single closed-loop system. The present work addresses this gap directly.

1.5 Research Questions

This dissertation is organized around three research questions:

RQ1: Can a closed-loop AI system driven by continuous Hexad profiling and Flow-state detection produce narrative that players perceive as more personalized than static, pre-authored narrative?

RQ2: Does PPO-driven narrative adaptation lead to measurably higher Flow experience (as captured by the GEQ Flow subscale) compared with a control condition that receives static narrative?

RQ3: How do players perceive and experience dynamically personalized narrative in terms of engagement and immersion?

RQ1 targets the subjective perception of personalization. RQ2 tests whether the objective psychological measure of Flow, operationalised through a validated instrument, differs between conditions. RQ3 is deliberately exploratory, intended to surface nuances that structured scales are liable to miss.

1.6 Research Aim and Objectives

The aim of this research is to show that a closed-loop, AI-driven narrative personalization system, packaged as a game-engine plugin, can measurably improve player experience when compared with static narrative delivery.

Six objectives support this aim:

1. Design and implement a five-module pipeline that integrates telemetry collection, player modelling, narrative generation, RL-based adaptation, and in-game content injection within a single coherent system.

2. Develop a method for deriving a six-dimensional Hexad player-type profile from behavioural telemetry, eliminating the need for a pre-play questionnaire.

3. Construct a Gymnasium-compatible MDP environment that frames narrative adaptation as a sequential decision problem with a Flow-based reward signal.

4. Build a RAG pipeline grounded in a structured game lore corpus that constrains LLM output to in-world content.

5. Design and conduct a between-subjects user study ($N = 50$) using validated psychometric instruments to evaluate the system's impact on Flow, immersion, and perceived personalization.

6. Contribute qualitative insight through reflexive thematic analysis of post-play interviews.

1.7 Significance of the Research

This research matters for three reasons. The first, and most visible, is integration. Each of the individual techniques involved (player modelling, RL-based adaptation, RAG-grounded LLM generation, episodic memory) has appeared in prior work. What has not appeared is a single deployable game plugin that wires all of them into a working closed loop.

The second reason is practical. Hexad profiling currently depends on a 24-item pre-play questionnaire, a one-time snapshot that cannot update during a session. Deriving Hexad scores directly from behavioural telemetry removes that friction entirely and allows the profile to evolve as the player's engagement patterns shift.

The third reason is conceptual. Reinforcement learning in games has been applied almost exclusively to mechanical difficulty adjustment, such as tuning enemy health pools, spawn rates, and resource drops. Using PPO for narrative action selection reframes the adaptation target: not how hard the game is, but what the story does next. That distinction opens a design space that prior work has left unexplored.

Beyond these contributions, the evaluation design, which combines validated psychometric instruments (GEQ, miniPXI) with qualitative thematic analysis, offers a reusable template for future studies of AI-generated game content. And because the entire system runs on locally hosted, open-weight models, it demonstrates that real-time narrative personalization does not require cloud infrastructure or commercial API access.

1.8 Scope of the Research

Several boundaries constrain the scope of this work. The testbed is a 3D RPG built on the AnyRPG Core framework, not a commercially released title; this ensures experimental control but limits ecological validity. The Hexad profiling method is rule-based and heuristic rather than a trained classifier. This is a deliberate choice driven by the cold-start constraint (no pre-existing player data), though it introduces approximation error relative to ground-truth questionnaire-derived types.

The LLM used is Phi-3.5 Mini (3.8B parameters), hosted locally via Ollama. This prioritises privacy and latency predictability over raw generation quality; larger cloud-hosted models would likely produce more fluent output but would bring dependency, cost, and data-governance concerns. The PPO agent is trained in a simulated environment rather than on human play data, so the learned policy reflects the simulator's transition dynamics rather than empirical player behaviour.

The user study is bounded to $N = 50$, which is sufficient for detecting medium-to-large effect sizes ($d \geq 0.5$) but not subtle ones. A between-subjects design eliminates carry-over effects at the cost of requiring careful condition matching for comparable baseline experience. Results are reported in Chapter 4 and discussed in Chapter 5.

Chapter 2

LITERATURE REVIEW

2.1 Chapter Overview

This chapter surveys the six bodies of research that underpin the system design: narrative personalization, player modelling, Flow theory, large language models for game narrative, reinforcement learning for adaptation, and episodic memory. For each area, the review identifies what prior work accomplished and, more importantly, where it stopped short, building incrementally toward the research gap that this dissertation sets out to fill.

Figure 2.1 maps the six domains and their inter-relationships. The solid arrows show how each domain feeds into the present work; the dashed cross-links highlight dependencies between domains that existing literature has acknowledged but not yet exploited within a single system.

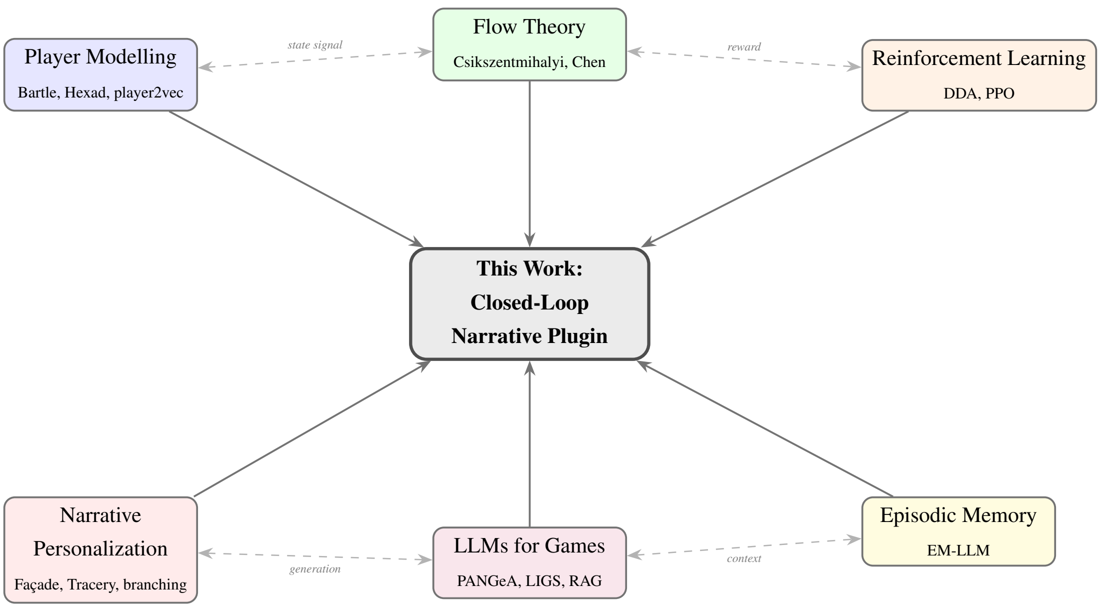


Figure 2.1: Literature map of reviewed research domains.

2.2 Background to the Problem

The question of how games might deliver individualized narrative experiences has occupied researchers for the better part of three decades, and the answers have evolved through several distinct paradigms, each solving part of the problem while leaving a conspicuous remainder.

The earliest serious attempts were structural. Drama management systems, exemplified by Mateas and Stern's Façade project (Mateas and Stern, 2003), sought to steer story structure toward dramatically satisfying outcomes by monitoring player actions in real time and intervening in the underlying plot. Façade was a landmark demonstration: it showed that an AI system could reshape a narrative arc around player behaviour within a single, self-contained scene. But the approach was brittle in ways that limited its influence on commercial game design. Every possible intervention had to be hand-authored in advance, and the system could not generalise to player actions that fell outside the authored space. Riedl and Bulitko (2013) surveyed the broader landscape of intelligent interactive narrative systems and diagnosed the underlying tension clearly: the more authorial control a system retains, the less it can accommodate genuine player agency, and vice versa. That tension has shaped nearly every subsequent design decision in the field.

A second wave of work approached the problem from a content generation perspective. Procedural narrative generation systems, including Tracery (Compton et al., 2015), the Expressionist framework (Kreminski et al., 2019), and their descendants, demonstrated that grammatical expansion rules could produce textual content that varied meaningfully across play-throughs. A player entering the same tavern twice might receive different descriptions, different rumours, different atmospheric detail. Yet these systems still depended on extensive hand-authored production rules, and they had no mechanism for sensing or responding to the player's psychological state. The content varied, but it varied randomly, with no model of what the player actually needed at that moment. Short (2019) crystallised the critique: procedural generation without a model of player intent risks producing content that is varied but meaningless, offering quantity of output without quality of relevance. A player who is struggling and a player who is thriving receive the same stochastic variety, which is no kind of personalization at all.

The commercial industry, meanwhile, converged on a middle path: narrative choice systems with pre-authored branches. Titles such as BioWare's *Mass Effect* trilogy, CD Projekt Red's *The Witcher* series, and ZA/UM's *Disco Elysium* offer the player decision points whose consequences ripple through the narrative, creating the impression that the story is shaped around the individual. The impression is powerful, and at its best it produces some of the most memorable moments in the medium. But the mechanism is ultimately deterministic: the narrative delivered at any point is a fixed function of the player's explicit choice history within a pre-specified decision space. These systems do not respond to behavioural signals (play pace, combat avoidance, exploration patterns, idle time) because they were never designed to observe them. The cost of extending authored branching is a further constraint. *Disco Elysium*, widely regarded as one of the most narratively ambitious RPGs ever released, contains approximately one million words of authored text, and yet it still delivers a structurally fixed narrative relative to the player's dialogue choices. Scaling this approach to accommodate implicit behavioural signals would multiply the authoring burden by orders of magnitude.

The most recent paradigm, and the one most relevant to the present work, uses generative AI to produce narrative content at runtime. PANGeA (Todd et al., 2024) demonstrated that LLMs can generate coherent, in-world narrative when grounded in a structured knowledge base delivered through the prompt, opening the possibility of adaptation that is both infinite in variety and anchored in the game's content universe. This is a qualitative shift: the narrative space is no longer bounded by what the writer pre-authored, but by what the model can generate within a grounding constraint.

The current work situates itself at the intersection of this generative paradigm and continuous player modelling. The narrative is not merely generated but *selected and styled* in response to a live, continuously updated model of the individual player's psychological state and behavioural profile, a combination that no prior system has achieved.

2.3 Theoretical and Conceptual Background

2.3.1 Player Modelling Approaches

Bartle Taxonomy and Its Limitations

The first systematic attempt to categorise players by what they actually want from a game, as opposed to what the game offers them, came from Bartle (1996), who proposed four types based on his observation of Multi-User Dungeon (MUD) communities in the 1980s: Achievers, who pursue goals and progression; Explorers, who want to map the world and understand its systems; Socializers, who value interaction with other players above all else; and Killers, who derive satisfaction from dominating others. The taxonomy's enduring contribution is the foundational insight that a single game contains, in effect, several different games (one for each type of player) and that design decisions that delight one type may alienate another.

That insight remains valuable, but the taxonomy itself has well-documented weaknesses. The most fundamental is that it is categorical: each player is slotted into one of four boxes, when the empirical evidence overwhelmingly suggests that players express multiple motivations simultaneously and in shifting proportions. Yee (2006), in a large-scale factor-analytic study of MMORPG players, showed that motivation is continuous and multi-dimensional, not discrete and singular. Beyond this, the Bartle taxonomy was derived from a narrow population (players of text-based MUDs in the 1980s and 1990s) and its applicability to the visual, mechanically complex, and often single-player games of the modern era is questionable at best. The four categories are neither mutually exclusive nor operationally defined in ways that permit reliable measurement. And the principal measurement instrument, the Bartle Test, is a forced-choice self-report questionnaire that captures a one-time snapshot coloured by social desirability bias rather than a dynamic, within-session picture of what the player is actually doing.

Hexad Typology

Tondello et al. (2016) set out to address these shortcomings by developing the Gamification User Types Hexad Scale, grounded explicitly in Self-Determination Theory (Ryan and Deci, 2000). The Hexad framework identifies six player types, each mapped to a core motivational driver: Achievers (mastery and progress), Explorers (discovery and curiosity), Socializers (social interaction and belonging), Free Spirits (autonomy and self-expression), Disruptors (challenging the status quo and testing boundaries), and Philanthropists (giving, helping, and altruism). Crucially, Hexad scores are continuous, measured on seven-point Likert items, and and a player can score highly on several dimensions at once. A player who is simultaneously a strong Explorer and a strong Philanthropist is not an anomaly under Hexad; it is the expected norm.

The original Hexad scale is a 24-item self-report questionnaire, validated for gamification contexts with acceptable internal consistency (Cronbach's α typically exceeding 0.70 per subscale). It has since been adopted across a range of application domains, including adaptive game design, recommendation systems, educational technology, and health interventions, establishing it as the de facto standard for research-grade player-type assessment (Hamari et al., 2014).

For the purposes of the present work, however, the Hexad scale carries a critical limitation: it is a *pre-play* questionnaire. A questionnaire-derived profile is a static snapshot, captured before the player touches the game, that cannot update as the player's engagement patterns evolve over the course of a session. A player who begins as a cautious Explorer and gradually shifts toward aggressive Achiever behaviour (a common trajectory in RPGs as players gain confidence) will be mislabelled for the entire session if the profile was frozen at minute zero. Moreover, administering a 24-item questionnaire before play introduces friction that degrades ecological validity and can prime participants to think explicitly about their motivations, altering the very behaviour the system needs to observe.

This dissertation addresses the limitation head-on by inferring Hexad-analogous continuous scores directly from behavioural telemetry, updating them continuously within the session. The mapping from telemetry signals to Hexad dimensions is a novel contribution, detailed in Section 3.4.

Data-Driven Player Modelling

A more recent line of work has applied deep learning to the player modelling problem. The player2vec approach (2024) uses Transformer-based self-supervised learning to encode sequences of in-game actions into dense, low-dimensional representations, much as word2vec encodes natural language tokens into embeddings that capture semantic relationships. The result is a learned embedding space in which players with similar behavioural trajectories cluster together, without any need for hand-labelled training data or predefined type categories. Zhu et al. (2023) surveyed the broader landscape of player modelling approaches for interactive narrative and identified the shift from questionnaire-based to behaviour-based modelling as perhaps the defining trend of the past decade.

The present work draws conceptual inspiration from player2vec, in particular the idea that a session-level embedding can capture the arc of a player's behaviour over time. But it implements a substantially simpler rule-based approach rather than a trained Transformer, for a pragmatic reason that the deep learning literature sometimes underemphasises. The cold-start problem is central: A Transformer-based session embedder needs a substantial corpus of historical play sequences (hundreds or thousands of sessions) to learn meaningful representations. For a first-session player using a newly deployed plugin with no prior data, that corpus does not exist. The rule-based Hexad profiler trades representational richness for immediate applicability: it begins producing useful profiles within the first few minutes of a session, before any learning-based method could have accumulated enough data to initialise.

More recently, Souza et al. (2025) explored a complementary direction: using LLMs themselves as player modelling engines, feeding behavioural summaries into a language model and prompting it to infer player traits. The approach is attractive because it leverages the same LLM infrastructure already present in a generative narrative pipeline, but it introduces its own limitations: inference latency scales with prompt length, the model's "understanding" of player psychology is at best a statistical reflection of its training corpus, and the approach has not yet been validated against ground-truth psychometric instruments. The present system avoids these issues by keeping player modelling as a lightweight, deterministic computation over raw telemetry, reserving the LLM exclusively for the narrative generation task where its generative capacity is genuinely needed.

2.3.2 Flow Theory in Game Design

Csikszentmihalyi's (1990) theory of Flow describes a psychological state of optimal experience, the kind of complete absorption in a challenging activity that makes a person lose track of time, forget self-consciousness, and perform at the upper edge of their ability. The theory's central claim is structural: Flow arises at the intersection of high perceived challenge and high perceived skill. When challenge substantially exceeds skill, the result is anxiety: the player is overwhelmed. When skill substantially exceeds challenge, the result is boredom: the player is under-stimulated. And when both challenge and skill are low, the result is apathy: the player is disengaged in the most concerning way, neither trying nor caring. Nakamura and Csikszentmihalyi (2002) later refined the model to emphasise that Flow is not a binary on/off state but a continuum influenced by attention, feedback loops, and the perceived meaningfulness of the activity.

The theory's influence on game design has been enormous, though often implicit rather than explicit. Chen's (2007) canonical analysis of game flow was the first to make the connection formal: he argued that many of the dynamic difficulty adjustment (DDA) mechanisms already present in commercial games (adaptive enemy scaling, hint systems, rubber-banding in racing games) are, in effect, Flow-optimization mechanisms operating without the theoretical label. The challenge–skill ratio has since become the standard operationalisation of Csikszentmihalyi's balance axis in computational models of player engagement.

What has received less attention, however, is the role of *narrative* in the Flow equation. DDA research has focused almost exclusively on mechanical parameters: how hard enemies hit, how fast they spawn, how generous the loot tables are. The assumption, rarely stated but pervasive, is that Flow is a function of mechanical challenge and mechanical skill. But players do not experience games as bare mechanical systems; they experience them as stories inhabited by characters in a world. A player who is mechanically in the Flow zone but narratively disengaged, receiving bland, generic descriptions while battling through a dungeon, may not report the subjective experience of Flow at all, because the narrative layer failed to sustain the sense of meaning that Flow requires.

The present work operationalises Csikszentmihalyi's model through a rule-based classifier that maps the `challenge_skill_ratio` (computed from behavioural telemetry) to one of four discrete states: FLOW (ratio in [0.75, 1.30]), ANXIETY (ratio $>$ 1.30), BOREDOM (ratio $<$ 0.75), and APATHY (a compound condition requiring high idle time, low dialogue engagement, *and* a low challenge–skill ratio). These thresholds are informed by prior work on computational Flow detection (Yannakakis and Hallam, 2009) and are formalised in Section 3.4. Crucially, the reward function of the PPO adaptation engine is shaped directly around these states, making Flow the central organising principle of the entire adaptation objective: not merely a diagnostic indicator, but the thing the system is explicitly trying to produce.

2.4 Review of Key Methods and Technologies

2.4.1 Large Language Models for Game Narrative

PANGeA (AAAI AIIDE 2024)

The most directly relevant antecedent for the narrative generation component of this system is Todd et al.'s (2024) Procedural Artificial Narrative using Generative AI (PANGeA). The system employs an LLM to generate narrative content at runtime, grounded in a structured game knowledge base that is delivered to the model through the prompt. PANGeA's core insight is both simple and powerful: LLMs *can* produce coherent, in-world narrative content, but only when the prompt supplies enough contextual grounding to anchor the model's generation within the fictional frame. Without that grounding, the model drifts: it invents characters that do not exist, references locations that contradict the map, or breaks the fourth wall by inserting contemporary idiom into a medieval setting.

PANGeA, however, stops short of personalisation. Its generation is procedural, driven by the current game state (what is happening in the world) rather than by the player's psychological state (what the player needs from the narrative at this moment). A player in Flow and a player teetering on the edge of boredom receive the same procedurally generated content, because PANGeA has no model of the player and no policy for selecting how to adapt the narrative's tone or direction. The current work extends the PANGeA paradigm by adding precisely those missing layers: a continuous player model and an RL-based adaptation policy that decides not just *what* to generate but *how* to modulate it.

Kumaran and Riedl (2025)

The LLM-Integrated Game Storytelling (LIGS) system, published at CHI 2025, approached the problem from the coherence angle. LIGS investigated how to integrate LLMs into game narrative pipelines while maintaining narrative consistency across extended play sessions, a problem that grows harder as sessions lengthen and the model must keep track of an expanding history of prior events, character states, and plot developments. LIGS introduced mechanisms for tracking narrative state across sessions and using that state to condition generation, anticipating the episodic memory approach that the current work develops independently.

A particularly valuable contribution of LIGS was its evaluation methodology. The study employed player experience questionnaires to compare AI-generated narrative against human-authored content and found that participants could not reliably distinguish the two when the LLM was appropriately constrained. This result matters for the present work: it provides empirical evidence that LLM-based narrative generation can be subjectively valid: players do not automatically perceive machine-generated text as inferior or artificial, provided the content stays within the boundaries of the game world.

Knowledge-Grounded and Quest-Level Generation

Two further lines of recent work have extended the scope of LLM-based narrative generation. Ashby et al. (2023) proposed a knowledge graph-grounded approach to personalised quest and dialogue generation in RPGs, using a structured game ontology to constrain LLM output in much the same way that RAG constrains it through retrieved passages. Their system generates quest content that is consistent with the game world's entity relationships, but it does not include a real-time player model or an adaptive policy: the personalisation is structural (what quests are offered) rather than tonal (how the narrative addresses the player's current psychological state).

Värtinen et al. (2024) demonstrated that GPT-class language models can generate complete RPG quest structures, including objectives, dialogue, and reward logic, that human evaluators rated as comparable in quality to hand-authored content. This finding reinforces the viability of LLM-based narrative generation for games and complements the LIGS result on player perception. However, like PANGeA and Ashby et al., the system generates quests procedurally from game state without modelling the player or adapting the narrative tone in response to engagement signals. The present work addresses this gap by combining generation quality (via RAG grounding) with adaptation intelligence (via PPO action selection conditioned on a continuous player model).

Prompt Grounding and Retrieval-Augmented Generation

The persistent challenge of keeping LLM outputs factually and fictionally consistent with a specific game world has driven the adoption of Retrieval-Augmented Generation (RAG; Lewis et al., 2020) in game narrative contexts. The idea is straightforward: rather than requiring the model to "know" the game's lore from its pre-training data (which, for a custom game world, it cannot possibly know), the system retrieves relevant passages from a curated knowledge base and injects them into the prompt at generation time. The model then generates within the semantic constraints established by the retrieved context, producing output that is grounded in the game's actual content rather than in the model's general-purpose parametric knowledge.

For game narrative, RAG offers a particularly attractive set of trade-offs. The lore corpus can be maintained, extended, and corrected independently of the model, meaning that world-building updates do not require retraining or fine-tuning. Retrieval ensures that only the passages most relevant to the current generation context are included in the prompt, avoiding the saturation of the model's finite context window. This last point is not a minor implementation detail: the Transformer architecture underlying modern LLMs (Vaswani et al., 2017) processes fixed-size context windows, and every token spent on irrelevant background is a token unavailable for the actual generation task. Efficient context usage is directly tied to generation quality.

The current system employs the sentence-transformers all-MiniLM-L6-v2 model (Wang et al., 2020) to embed both the lore corpus and generation queries into a 384-dimensional semantic space. Retrieval operates via top-k cosine similarity ($k = 3$), returning the three most relevant lore passages for inclusion in the prompt. The small model footprint (roughly 80 MB) and sub-millisecond retrieval latency make this approach well suited to real-time game narrative, where the generation pipeline must complete within the player's tolerance for loading delay.

2.4.2 Reinforcement Learning for Game Adaptation

Dynamic Difficulty Adjustment

Dynamic Difficulty Adjustment (DDA) has been an active research area since Hunicke and Chapman (2004) demonstrated that adaptive difficulty could measurably improve player retention in *Super Mario Bros*. The core idea is intuitive: if a player is dying too often, make the game easier; if the player is breezing through unchallenged, make it harder. Computational approaches to DDA have employed rule-based controllers, Bayesian models, and more recently neural networks to infer an appropriate challenge level from behavioural signals and adjust game parameters accordingly (Lopes and Bidarra, 2011). The predominant formulation treats DDA as a feedback control problem: the challenge level is the controlled variable, the player's observed engagement state is the feedback signal, and the controller minimises the deviation from a target state, typically some proxy for Flow.

DDA works, and it has been deployed in commercial games ranging from racing titles (rubber-banding AI) to action games (adaptive enemy health pools). Climent et al. (2024) recently formalised a general framework for designing RL-based DDA agents in single-player action games, demonstrating that PPO and similar policy gradient methods can learn effective difficulty policies directly from player interaction data. Their framework validates the use of RL for real-time game adaptation and provides design patterns (state representation, reward shaping, training regimes) that informed the present work's MDP formulation. But the scope of what DDA adapts remains narrow. DDA adjusts *mechanical* parameters: how much damage enemies deal, how many spawn per wave, how generous the loot tables are. It does not adjust what the *story does*. A player who is mechanically in the Flow zone but narratively disengaged, trudging through a dungeon with no sense of purpose because the quest text failed to land, is invisible to a DDA system.

The present work extends the DDA paradigm in two ways that, taken together, amount to a reframing. First, the adaptation target is not mechanical difficulty but *narrative content*: the tone, urgency, mystery, and emotional register of the story the player receives. Second, the adaptation mechanism is a learned policy (PPO) rather than a hand-tuned feedback controller. These extensions shift the problem from mechanical tuning, which is essentially an engineering exercise, into a narrative and psychological design space where the "right answer" depends on who the player is and what kind of experience they are seeking.

PPO and Stable-Baselines3

Proximal Policy Optimization (PPO; Schulman et al., 2017) is a policy gradient reinforcement learning algorithm that has become the workhorse of modern RL research, owing to a combination of relative simplicity, reliable performance across diverse environments, and a clipped surrogate objective that prevents the destructively large policy updates which plagued earlier policy gradient methods. Where algorithms like TRPO imposed a hard constraint on the KL divergence between successive policies, PPO achieves a similar stabilising effect through a simpler clipping mechanism that is both easier to implement and cheaper to compute. The result is an algorithm that trains reliably with minimal hyperparameter tuning, a practical advantage that has made PPO the default choice for applications ranging from robotics to game-playing agents (Mnih et al., 2015).

Stable-Baselines3 (Raffin et al., 2021) provides production-quality PyTorch implementations of PPO and other standard RL algorithms, with interfaces compatible with the Gymnasium (formerly OpenAI Gym; Brockman et al., 2016) environment specification. The library handles the engineering details (vectorised environments, gradient accumulation, logging) so that the researcher can focus on environment and reward design. The current work employs Stable-Baselines3 v2.x with the MlpPolicy architecture (two hidden layers of 64 units each), training over 200,000 timesteps in a custom Gymnasium environment that simulates narrative adaptation episodes. The use of a well-tested, reproducible RL implementation is particularly important for a system that will be evaluated on human experience: if the policy behaves unpredictably during the user study, the evaluation is compromised regardless of how well the rest of the pipeline performs.

2.4.3 Episodic Memory for LLMs

One of the most persistent limitations of Transformer-based language models is their fixed context window: the model can attend to a bounded number of tokens, and anything that falls outside that window is, for all practical purposes, forgotten. In a game narrative context, this means that an LLM generating a scene late in a play session has no access to the events it narrated twenty minutes earlier, unless those events are explicitly reintroduced into the prompt. Without some form of memory, the generated narrative will be locally coherent but globally amnesiac, referencing characters and situations that have no connection to what the player has already experienced.

Fountas et al. (2024) introduced EM-LLM, an architecture that addresses this problem by endowing LLMs with a form of human-like episodic memory. The system partitions a long context stream into discrete memory events, scores each event by importance and surprise, and selectively retrieves the most relevant events for inclusion in future prompts. The result is a kind of principled compression: rather than attempting to cram the entire interaction history into a fixed-size window, the model recalls only the events most salient to the current generation context. This mirrors, at least in broad strokes, the way human episodic memory works — not as a complete recording but as a selective reconstruction weighted toward events that were surprising, emotional, or consequential.

The current work incorporates a lightweight episodic memory inspired by EM-LLM, simplified to match the scale of the problem. The `EpisodicMemory` class maintains a capped list of up to ten narrative events, each tagged with an importance score in [0, 1] and a timestamp. When the list exceeds capacity, it is sorted by importance in descending order and truncated; the least important event is evicted, preserving the narrative moments that matter most. At generation time, the top five events by importance are retrieved and formatted as a context block in the LLM prompt, providing the model with a compressed narrative history that enables references to past events, maintains character continuity, and prevents the kind of temporal inconsistency that would break immersion.

The simplification relative to EM-LLM's full architecture (which includes surprise-based segmentation, hierarchical memory, and retrieval scoring informed by both recency and salience) is deliberate and appropriate. A 30-minute game session produces approximately 15–20 narrative events, a volume easily managed by a flat list with importance-weighted eviction. The overhead of a full hierarchical memory system would add complexity without proportional benefit at this session scale.

2.5 Related Work

Table 2.1 draws together the key related works and maps their coverage across the six dimensions that define the scope of this dissertation. The pattern is immediately visible: each prior system addresses one or two dimensions well, but none covers more than three, and none closes the loop from player observation through adaptive generation back to in-game delivery.

Table 2.1: Summary of related work on narrative personalization and player modelling.

System	**Reference**	**Player Model**	**Flow Reward**	**LLM Gen.**	**RAG**	**Episodic Memory**	**Complete Loop**
PANGeA	Todd et al. (2024)	No	No	Yes	Partial	No	No
LIGS	Kumaran and Riedl (2025)	No	No	Yes	No	Partial	No
KG-Quest	Ashby et al. (2023)	No	No	Yes	KG	No	No
GPT-Quests	Värtinen et al. (2024)	No	No	Yes	No	No	No
DDA	Hunicke and Chapman (2004)	Behav.	Implicit	No	No	No	No
RL-DDA	Climent et al. (2024)	Behav.	Implicit	No	No	No	No
LLM-PM	Souza et al. (2025)	LLM	No	No	No	No	No
player2vec	Devlin et al. (2024)	Deep	No	No	No	No	No
EM-LLM	Fountas et al. (2024)	No	No	Yes	No	Yes	No
This work		**Hexad+ Flow**	**Explicit**	**Yes (RAG)**	**Yes**	**Yes**	**Yes**

The gap is not that these individual techniques are unknown; each has been demonstrated in isolation. The gap is that no one has wired them together. Continuous Hexad profiling derived from behavioural telemetry, Flow-based RL reward shaping, RAG-grounded LLM generation, and episodic session memory have never been integrated into a single closed-loop system that runs in real time within a game engine. The present work addresses that gap by combining all six components into a deployable Unity plugin and evaluating the result through a mixed-methods study that measures both the psychological impact (via the GEQ and miniPXI) and the subjective perception (via the NPS-3 and semi-structured interviews).

Chapter 3

METHODOLOGY

3.1 Chapter Overview

This chapter lays out the research design, walks through each of the five system modules in detail, describes the testbed game, documents the implementation, and specifies the evaluation plan. The methodology follows a mixed-methods approach (Creswell and Plano Clark, 2018), combining a controlled between-subjects experiment with qualitative interview analysis to capture both the measurable and the experiential dimensions of narrative personalization.

3.2 Research Methodology

The research adopts a mixed-methods design that pairs a controlled between-subjects experiment with a qualitative interview study. The experimental component is designed for causal inference: if a significant difference in player experience is observed between conditions, the controlled design allows it to be attributed to the adaptive narrative system rather than to confounding factors. The qualitative component captures something the numbers cannot: the texture of how players actually experience personalization, what they notice, what they value, and what fails to land.

A between-subjects design was chosen over a within-subjects alternative to eliminate order effects and carry-over contamination. The concern is straightforward: a player who first experiences adaptive narrative and then switches to static narrative (or vice versa) would inevitably bring expectations, comparisons, and memory from the first condition into the second. The experience of the second condition would be coloured by the first, and any difference between conditions would be confounded with the order in which they were encountered. With a between-subjects design, each participant experiences exactly one condition, cleanly eliminating this contamination at the cost of requiring a larger sample and careful matching to ensure the two groups are comparable at baseline.

The experimental condition (Experimental) deploys the full five-module AI system described in this chapter. The control condition (Control) replaces the Narrative Generation Service and the Adaptation Engine with a `StaticNarrativeContent` ScriptableObject, a pre-authored collection of narrative fragments delivered in response to the same game triggers, in the same visual format, but with no dependence on player state. Everything else (the game environment, the objectives, the NPC placement, the session duration, the UI layout) is identical across conditions. This isolation is critical: if the Experimental group differs from the Control group on the dependent measures, the difference can be attributed to the adaptive narrative system itself, not to differences in game content, visual presentation, or session structure.

3.3 System Architecture

The system is organized as five modules in a closed-loop pipeline. Figure 3.1 describes the architecture and the data flows between components. The five modules (telemetry collection, player modelling, PPO-based adaptation, RAG-grounded narrative generation, and content injection) form a pipeline from player behaviour back to in-game delivery; dashed boxes in the diagram indicate platform boundaries.

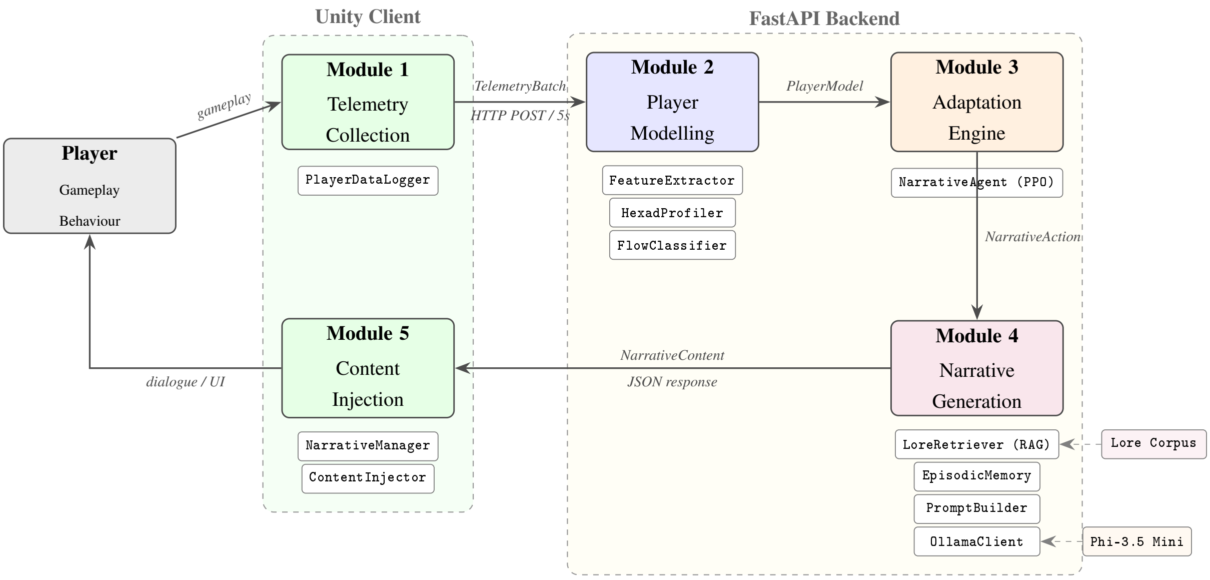


Figure 3.1: Closed-loop system architecture.

The pipeline operates as a continuous loop, cycling once per telemetry window. The Unity `PlayerDataLogger` aggregates behavioural signals over five-second windows and transmits a `TelemetryBatch` to the FastAPI backend via HTTP POST (with WebSocket support available for lower-latency streaming when network conditions permit). On the backend, the Player Modelling Service processes each incoming batch through three sequential stages: the `FeatureExtractor` computes eleven normalised features from the raw telemetry, the `HexadProfiler` derives the six-dimensional continuous Hexad profile from the cumulative session history, and the `FlowClassifier` assigns a Flow state based on the most recent data. The assembled `PlayerModel` is then passed to the `NarrativeAgent`, which selects a `NarrativeAction`, using the cold-start heuristic during the first 120 seconds and the trained PPO policy thereafter. The Narrative Generation Service takes the selected action and uses it to formulate a query for the `LoreRetriever` (RAG), consults the `EpisodicMemory` for relevant prior narrative events, assembles the full two-message prompt through the `PromptBuilder`, and dispatches a generation request to the locally hosted Ollama LLM. The resulting `NarrativeContent` (a JSON payload containing the generated text, content type, and speaker attribution) is returned to the Unity client and injected at the appropriate in-game display point by the `ContentInjector`.

3.4 Player Modelling Module

The player modelling module is responsible for answering two questions about the player at every point in the session: *what kind of player are they?* (the Hexad profile) and *how are they doing right now?* (the Flow state). Everything downstream (the action the PPO agent selects, the tone the LLM adopts, the content the player ultimately sees) depends on the accuracy and responsiveness of these two answers.

3.4.1 Telemetry Collection

The telemetry system captures 23 behavioural fields per five-second window, providing the raw signal from which both the Hexad profile and the Flow state are derived. Table 3.1 lists the complete schema.

Table 3.1: TelemetryBatch schema (23 fields).

Field	**Type**	**Description**
player_id	str	Unique player identifier
session_id	str	Unique session identifier
window_start	float	Window start time (Unix timestamp)
window_end	float	Window end time (Unix timestamp)
kills	int	Enemy kills in window
deaths	int	Player deaths in window
damage_taken	float	Total damage received
damage_dealt	float	Total damage inflicted
abilities_used	int	Ability activations
abilities_hit	int	Abilities that connected
objectives_completed	int	Objectives completed
objectives_attempted	int	Objectives attempted
tiles_explored	int	Map tiles visited
total_tiles	int	Total traversable tiles (default 100)
npc_interactions_voluntary	int	Voluntary (unsolicited) NPC interactions
dialogue_lines_shown	int	Dialogue lines displayed
dialogue_lines_skipped	int	Dialogue lines skipped by player
lore_items_read	int	Lore collectibles read
lore_items_found	int	Lore collectibles discovered
idle_seconds	float	Seconds with no player input
session_elapsed	float	Total elapsed session time (seconds)

Field	Type	Description
current_location	str	Scene/zone identifier
current_objective	str	Active quest objective identifier

The five-second window granularity represents a deliberate trade-off between responsiveness and noise sensitivity. A shorter window would allow faster adaptation cycles but would amplify transient spikes: a single unlucky death or a momentary pause to answer a phone call would distort the player model. A longer window would produce smoother signals but at the cost of delayed response: if a player tips from Flow into anxiety, the system should notice within seconds, not minutes. Five seconds proved to be a practical sweet spot during development. In practice, the system further smooths the signal by accumulating a sliding window of recent batches (typically six, spanning approximately 30 seconds) before computing the feature vector, allowing multi-window averages to absorb transient fluctuations.

3.4.2 Feature Extraction (11 Features)

The `FeatureExtractor` aggregates one or more `TelemetryBatch` records into eleven normalized scalar features, each clipped to [0, 1] (or [0, 3] for the `challenge_skill_ratio`). Table 3.2 presents the computation formulae.

Table 3.2: Eleven extracted features and computation formulae.

Feature	Formula	Notes
kd_ratio	clip(kills / max(deaths, 1) / 5.0, 0, 1)	Norm. at 5:1 K/D
objective_rate	clip(obj_comp / max(obj_att, 1), 0, 1)	Completion proportion
explore_rate	clip(tiles_explored / total_tiles, 0, 1)	Map coverage
dialogue_engage	clip(1 − skipped / max(shown, 1), 0, 1)	1 = reads all
ability_accuracy	clip(hit / max(used, 1), 0, 1)	Combat precision
damage_taken_norm	clip(dmg / (elapsed_min + 1) / 200, 0, 1)	DPM normalized
idle_ratio	clip(idle_s / elapsed, 0, 1)	Fraction idle
lore_engage_rate	clip(read / max(found, 1), 0, 1)	Lore depth
skill_score	0.35 kd + 0.30 obj + 0.20 acc + 0.15(1−idle)	Composite skill
challenge_score	0.50 dmg + 0.30(1−obj) + 0.20 death_ratio	Composite challenge
challenge_skill_ratio	clip(challenge / max(skill, 10^{-6}), 0, 3)	Core Flow signal

The `skill_score` composite weights combat effectiveness (`kd_ratio`, `ability_accuracy`) most heavily, followed by objective completion and attention (inverse `idle_ratio`). The `challenge_score` composite weights damage pressure most heavily, followed by objective failure rate and death rate. Dividing challenge by skill yields the `challenge_skill_ratio`, which operationalizes Csikszentmihalyi's (1990) challenge–skill balance axis.

3.4.3 Hexad Profile Computation (6 Dimensions)

The `HexadProfiler` computes six continuous scores in [0, 1] from accumulated telemetry, each representing the strength of a Hexad player-type signal. Table 3.3 shows the computation weights.

Table 3.3: Hexad dimension computation weights from telemetry signals.

Dimension	**Formula**	**Interpretation**
Achiever	$0.70 \times$ obj_rate + $0.30 \times$ norm(kills, 20)	Objective + combat
Explorer	$0.50 \times$ explore_rate + $0.50 \times$ lore_rate	Map + lore coverage
Socializer	$0.60 \times$ norm(vol_npc, 10) + $0.40 \times$ dialogue	Seeks NPCs + reads
Free Spirit	$0.50 \times$ explore_rate + $0.50 \times (1 -$ obj_rate)	Explores off-path
Disruptor	$0.65 \times$ norm(kills, 20) + $0.15 \times$ norm(deaths, 10) + $0.20 \times$ norm(dmg, 1000)	Aggressive combat
Philanthropist	$0.60 \times$ dialogue_engage + $0.40 \times$ lore_rate	Narrative engagement

The mapping from behavioural signals to Hexad dimensions was established through a principled alignment process guided by the original Hexad motivation descriptions (Tondello et al., 2016). The logic proceeds dimension by dimension. Achievers seek mastery and progress, so the signal proxies are objective completion rate and combat kill count, both direct indicators of goal-oriented play. Explorers seek discovery, so map coverage and lore engagement serve as proxies: a player who visits every corner of the map and reads every collectible is behaving as an Explorer regardless of what a questionnaire might say. Socializers seek connection, so voluntary NPC interaction count and dialogue engagement rate are weighted; a player who seeks out NPCs unprompted and reads their dialogue is exhibiting Socializer behaviour. Free Spirits resist imposed structure, so the proxy is exploration *uncorrelated with* objective completion, wandering off the critical path rather than following it. Disruptors challenge systems, captured by aggressive combat behaviour (high kill counts, high damage dealt, willingness to die repeatedly). Philanthropists give and share, captured by depth of engagement with narrative and lore content, a proxy for the motivation to understand and connect with the game world rather than merely to conquer it.

All profiles are initialised to 0.5 (the neutral midpoint) in the absence of data, preventing cold-start artefacts from biasing the narrative action selection before the system has observed enough behaviour to form a meaningful profile.

3.4.4 Flow State Classification (MDP States)

The `FlowClassifier` implements a decision-tree classifier over the `challenge_skill_ratio`, `idle_ratio`, and `dialogue_engage_rate` features. Table 3.4 formalizes the rules.

Table 3.4: Flow state classification rules (evaluated in priority order).

Priority	**Condition**	**State**	**Confidence**
1 (highest)	idle $>$ 0.40 AND dialogue $<$ 0.30 AND ratio $<$ 0.60	APATHY	0.70
2	ratio $<$ 0.75	BOREDOM	$0.60 + (0.75 - \text{ratio}) \times 0.80$
3	ratio $>$ 1.30	ANXIETY	$0.60 + (\text{ratio} - 1.30) \times 0.40$
4 (default)	$0.75 \leq$ ratio ≤ 1.30	FLOW	$\max(0.55, 1.0 - \lvert\text{ratio} - 1.0\rvert \times 1.2)$

Figure 3.2 visualises the classification logic as a decision tree. Conditions are evaluated in priority order (top to bottom); the first satisfied condition determines the classification.

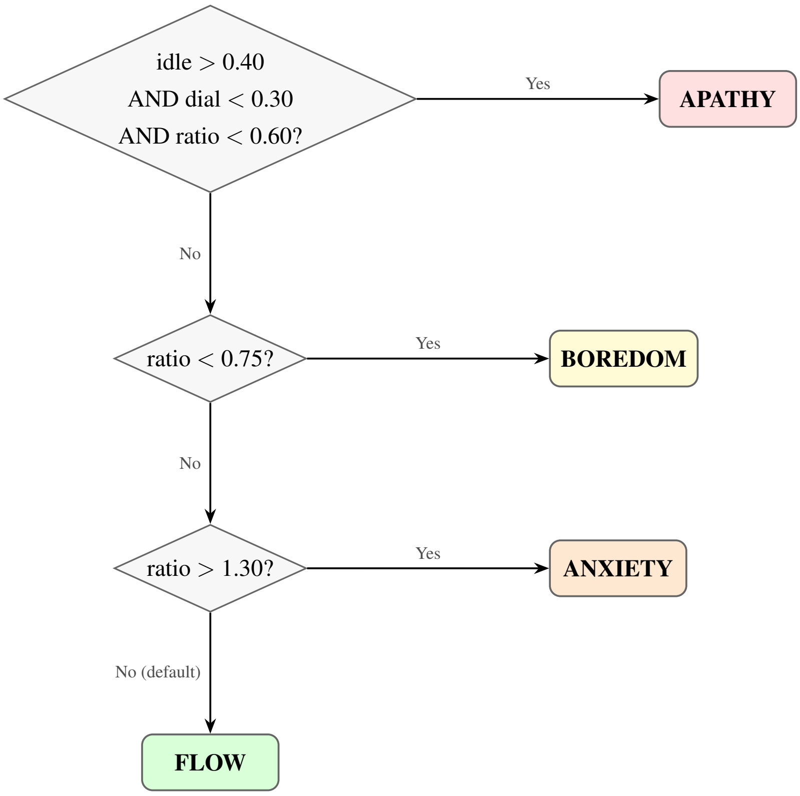


Figure 3.2: Flow state classification decision tree.

The priority ordering matters. APATHY is evaluated first because it represents a qualitatively distinct disengagement pattern: the player has not merely lost interest in the challenge (which would be BOREDOM) but has stopped meaningfully interacting with the game altogether. Without the compound condition check, such a player would be classified as BORED on the challenge–skill ratio alone, and the system would respond with urgency-boosting narrative. But a disengaged player does not need urgency; they need a reason to care. The APATHY classification ensures the system responds with mystery and intrigue rather than pressure.

The confidence scoring for BOREDOM and ANXIETY is proportional to the distance from the FLOW band boundary: a player with a challenge–skill ratio of 0.40 is classified with higher confidence than one at 0.70, because the former is further from the zone of optimal engagement. FLOW confidence is highest when the ratio sits closest to 1.0, the theoretical point of perfect challenge–skill balance, and tapers as the ratio approaches either boundary.

3.5 Narrative Generation Module

The narrative generation module is the part of the system that actually writes the story. Everything upstream (the telemetry, the feature extraction, the Flow classification, the Hexad profiling, the PPO action selection) exists to inform this module about *what kind of narrative to produce*. The module itself is responsible for producing it: grounded in the game's lore, consistent with the session's narrative history, and formatted for injection into the game's UI.

3.5.1 RAG Pipeline

The `LoreRetriever` implements in-memory semantic search using the sentence-transformers all-MiniLM-L6-v2 model, which produces 384-dimensional L2-normalised embeddings. The lore corpus, comprising the world, character, and quest lore documents stored in the `lore/` directory, is pre-indexed at server startup. Each document is chunked by paragraph (double-newline delimiters), and each chunk is embedded into the 384-dimensional space. At retrieval time, the query (constructed from the player's current Flow state, activity, and location) is embedded and cosine similarities are computed against all stored chunk embeddings via a dot product (since all vectors are unit-normalised, cosine similarity reduces to the dot product). The top $k = 3$ chunks by similarity are returned and formatted as bullet points in the LLM user prompt, providing the model with the lore context most relevant to the current generation task.

This design was chosen over a persistent vector database (e.g., ChromaDB, Pinecone) for two reasons that proved decisive during development. First, ChromaDB's Pydantic v1 dependency created compatibility issues with the project's Python 3.14 runtime that could not be cleanly resolved. Second, and more fundamentally, the lore corpus is small (three files, approximately 30 paragraphs, a few hundred embeddings at most), which renders the indexing, persistence, and query optimisation machinery of a full vector database unnecessary overhead. In-memory retrieval at this corpus scale completes in sub-millisecond time, well within the latency budget of the narrative generation pipeline.

3.5.2 Episodic Session Memory

The `EpisodicMemory` class maintains a capped list of `MemoryEvent` records, each comprising a text description, an importance score in [0, 1], and a timestamp. When the list exceeds the maximum capacity of ten events, it is sorted by importance in descending order and truncated to the top ten. This importance-weighted eviction ensures that the most narratively significant events are retained even as minor events accumulate.

At prompt construction time, the top five events by importance score are selected and formatted as a context block, providing the LLM with a compressed narrative history that enables references to past events and prevents temporal inconsistency.

3.5.3 Dynamic Prompt Construction

The `PromptBuilder` assembles a two-message prompt (system + user) for the Ollama API. The system prompt encodes the LLM's role (narrative writer for the game world), output length constraints (2–4 sentences for dialogue, 1–2 for descriptions), format requirements (valid JSON only, specific schema), and the emotional tone mapped from the current Flow state:

- ANXIETY: "urgent and supportive — the player is struggling"
- BOREDOM: "exciting and surprising — inject energy"
- FLOW: "immersive and atmospheric — maintain the experience"
- APATHY: "intriguing and mysterious — re-engage the player"

The user prompt injects: the player's current Flow state and challenge–skill ratio; the current activity (IN_COMBAT, EXPLORING, IN_DIALOGUE, IDLE); session elapsed time; the Hexad profile percentages; the selected NarrativeAction with definitions; the current location and active quest stage; the three retrieved lore chunks; the episodic memory context; and explicit content constraints (no real-world references, no anachronisms, no fourth-wall breaks, character voice descriptions).

3.5.4 LLM Integration (Ollama / Phi-3.5 Mini)

The `OllamaClient` sends the assembled prompt to a locally hosted Ollama instance using the `/api/chat` endpoint. The primary model is Phi-3.5 Mini (Microsoft, 2024) which has 3.8B parameters, with Llama 3.2 3B as a configured fallback. The generation parameters are: temperature 0.75, top_p 0.90, maximum tokens 200, JSON format mode enabled. The 8-second HTTP timeout ensures that a slow or stalled generation does not block the main game loop.

When the LLM response is malformed JSON, contains a fallback signal, or the request times out, the `OllamaClient` returns a pre-authored fallback string corresponding to the requested `NarrativeAction`. These fallback strings are intentionally minimal (a single sentence of tonally appropriate narrative) designed to maintain the illusion of a functioning narrative system without exposing the player to error messages, empty dialogue boxes, or other artefacts that would break immersion. The player sees a brief, contextually appropriate line; they do not see that the LLM failed to produce valid output. This graceful degradation is essential for a system that will be used in a controlled user study, where a single visible error could contaminate a participant's experience and compromise the evaluation.

3.6 Adaptation Engine (MDP + PPO)

The adaptation engine is the decision-making core of the system: given what the player modelling module knows about the player right now, what should the narrative do next? This question is formalised as a reinforcement learning problem, which allows the answer to be *learned* from experience rather than hand-coded by the designer.

3.6.1 MDP Formulation

The narrative adaptation problem is formalised as a Markov Decision Process (S, A, T, R, γ) implemented as a Gymnasium environment.

State space S: A Box observation space of shape (7,) with all values in [0, 1]. Table 3.5 defines the seven dimensions.

Table 3.5: MDP observation vector (7 dimensions).

Index	Signal	Description
0	flow_score	Encoded Flow state (FLOW=1.0, BOREDOM=0.3, ANXIETY=0.2, APATHY=0.1)
1	csr_norm	challenge_skill_ratio / 3.0
2	hexad_explorer	Explorer dimension score
3	hexad_socializer	Socializer dimension score
4	hexad_achiever	Achiever dimension score
5	hexad_disruptor	Disruptor dimension score
6	session_progress	session_elapsed / 1800.0 (normalized to 30-min max)

Action space A: A Discrete(8) action space corresponding to the eight `NarrativeAction` values. Table 3.6 summarizes the actions.

Table 3.6: Eight narrative actions with descriptions and intended effects.

Idx	Action	Intended Effect
0	LOWER_STAKES	Reduce narrative tension; provide respite
1	RAISE_STAKES	Increase narrative urgency and weight
2	ADD_MYSTERY	Introduce a curiosity hook or unexplained element
3	ADD_HUMOR	Provide comic relief to reduce tension
4	PROVIDE_GUIDANCE	Give struggling player directional narrative support
5	INCREASE_URGENCY	Re-energize a bored or disengaged player
6	LORE_REWARD	Deliver a lore discovery payoff for curious players
7	NO_CHANGE	Suppress narrative injection; maintain current experience

Transition dynamics T: The simulated training environment models stochastic Flow state transitions. When the agent selects a "helpful" action for the current state (e.g., ANXIETY $\rightarrow$ $\{$PROVIDE_GUIDANCE, LOWER_STAKES$\}$), the environment transitions to FLOW with probability 0.75 or remains in the current state with probability 0.25. This stochasticity prevents the policy from memorizing a deterministic mapping and encourages generalization across states.

Figure 3.3 visualises the state transitions and the narrative actions that drive them. Solid arrows show transitions caused by helpful narrative actions ($p = 0.75$ to FLOW), self-loops indicate remaining in the current state, and the dashed arrow shows the stochastic drift from FLOW to BOREDOM when no helpful action is needed.

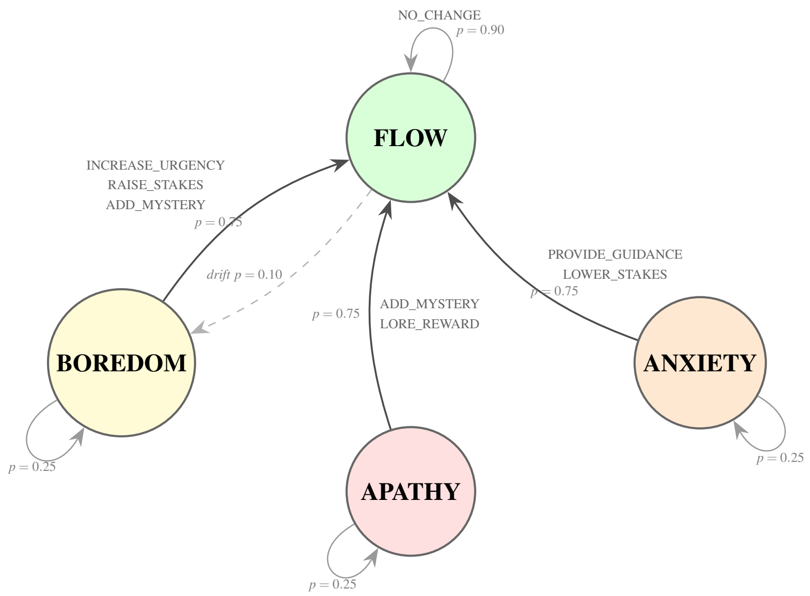


Figure 3.3: MDP state-transition diagram.

Discount factor γ: 0.95, reflecting moderate preference for near-term reward while maintaining long-horizon planning.

3.6.2 Reward Function Design

The reward function is defined over (prev_state, new_state, action, steps_in_state, last_3_actions) and comprises five components, detailed in Table 3.7.

Table 3.7: Reward function components and values.

Component	**Value**	**Trigger**
State reward (FLOW)	$+1.0$	new_state == FLOW
State reward (BOREDOM)	-0.3	new_state == BOREDOM
State reward (ANXIETY)	-0.5	new_state == ANXIETY
State reward (APATHY)	-0.8	new_state == APATHY
Step penalty	-0.05	Every step
Flow streak bonus	$+0.20$	FLOW AND steps_in_state > 2
Transition bonus	$+0.30$	prev $\in$ {ANXIETY, BOREDOM} AND new == FLOW
Repetition penalty	-0.15	All 3 recent actions identical

The reward function encodes a clear priority ordering. APATHY draws the heaviest penalty (-0.8) because it represents the deepest disengagement, a player who has essentially given up. ANXIETY (-0.5) is penalised more than BOREDOM (-0.3) because an overwhelmed player is at higher risk of quitting the session entirely. The step penalty (-0.05) applies at every timestep regardless of state, creating constant pressure toward action efficiency and discouraging the agent from drifting into passive NO_CHANGE loops. The Flow streak bonus ($+0.20$) rewards sustained optimal experience, not just momentary hits of Flow: it activates only after the player has been in FLOW for more than two consecutive steps, encouraging the agent to maintain conditions rather than simply reaching them. The transition bonus ($+0.30$) provides additional shaping reward for the key recovery transitions, moving a player from ANXIETY or BOREDOM back into FLOW, which are the most valuable moments for the system to get right. Finally, the repetition penalty (-0.15) fires when all three most recent actions are identical, discouraging degenerate policies in which the agent converges on a single favourite action and applies it regardless of context.

3.6.3 Cold-Start Heuristic

For the first 120 seconds of a session, or when the PPO model is unavailable, the `NarrativeAgent` falls back to a deterministic heuristic policy:

- ANXIETY $\rightarrow$ PROVIDE_GUIDANCE
- BOREDOM $\rightarrow$ INCREASE_URGENCY
- APATHY $\rightarrow$ ADD_MYSTERY

- FLOW → NO_CHANGE

The 120-second threshold serves a dual purpose. It gives the player time to settle into the game world (learning the controls, orienting to the environment, forming initial goals) before the learned policy begins making adaptation decisions. And it gives the player modelling module time to accumulate enough telemetry for the Hexad profile and Flow classification to be meaningful. A PPO policy operating on a player model derived from ten seconds of data would be making decisions on noise; the heuristic ensures that the first two minutes of narrative adaptation are sensible even when the signal is thin.

3.7 Unity Plugin Design

The Unity plugin comprises three C# components: `PlayerDataLogger`, `NarrativeManager`, and `ContentInjector`.

`PlayerDataLogger` is a MonoBehaviour that maintains running accumulators for all 23 telemetry fields. Every five seconds, it serializes the accumulated values into a `TelemetryBatch` JSON object and dispatches it to the FastAPI backend via an async HTTP POST. It exposes a public API for game systems to register events: `RecordKill()`, `RecordDeath()`, `RecordDialogueShown()`, and others. This event-based registration model decouples the telemetry system from specific game mechanics, making the plugin applicable to any Unity game that exposes these event hooks.

`NarrativeManager` polls the `/narrative/generate` endpoint following each telemetry transmission. `ContentInjector` maps `NarrativeContentType` to the appropriate UI mechanism: dialogue content is routed to the DialoguePanel, description content to the DescriptionPanel, and quest updates to the QuestLog. The injector implements a content queue with a 3-second minimum spacing between consecutive injections to prevent narrative overcrowding.

For the Control condition, the `NarrativeManager` is replaced by a `StaticNarrativeManager` that reads from a `StaticNarrativeContent` ScriptableObject and fires narrative events at scripted triggers with no player-state dependency.

3.8 Testbed Game Design

The testbed is a 3D RPG built in Unity 6 LTS using the AnyRPG Core framework (AnyRPG, 2024). AnyRPG provides a complete RPG engine (combat, questing, inventory, NPC dialogue, and character progression) out of the box, which allowed the development effort to focus on the narrative personalization system rather than on reimplementing standard RPG mechanics from scratch.

The game world is set in the Contested Vale, a once-fertile trade corridor now fought over by two rival factions. The setting was chosen to support a range of player behaviours: combat encounters for Achievers and Disruptors, exploration opportunities for Explorers and Free

Spirits, NPC interactions for Socializers, and lore collectibles for Philanthropists. The world is described in structured lore documents (stored in the `lore/` directory) that double as the RAG corpus for the narrative generation pipeline.

Three navigable areas make up the game world:

1. **Commander Varis's Post**, a fortified camp at the northern edge serving as the starting area. Varis is the primary quest giver: a methodical, unsentimental military commander who gives orders without explaining them. Players who engage extensively with Varis register high Achiever signals.

2. **The Contested Zone**, a combat-focused area containing enemy units and environmental hazards. Players who clear this area efficiently register high Achiever and Disruptor signals; players who explore its periphery register Explorer signals.

3. **The Waystone / Chronicler's Camp**, a neutral area where the Chronicler, a scholar who has observed the Vale for thirty years, provides lore and context. The presence of lore collectibles creates a natural Explorer/Philanthropist signal divergence.

Three NPCs populate the world, each designed with a distinct voice and function that the LLM must maintain when generating dialogue for them. Commander Varis is the primary quest giver: a methodical, unsentimental military officer who gives orders and expects compliance. The Chronicler is the lore source: a scholar who has observed the Vale for thirty years and speaks in measured, academic prose. Sera the Merchant is the pragmatic intermediary: a supply trader whose dialogue is clipped, transactional, and occasionally sardonic. These three voices serve as a consistency test for the narrative generation pipeline; if the LLM generates dialogue for the Chronicler that sounds like Varis, the player will notice.

The main quest (Clear the Advance) provides a linear spine through the combat zone, ensuring that all players encounter a minimum threshold of challenge and narrative content. A secondary collection quest (Salvage the Fallen) encourages off-path exploration and rewards lore engagement. A standard play session runs approximately 25–30 minutes, which is long enough for the player modelling module to build a meaningful profile but short enough to remain practical within the constraints of a controlled user study.

3.9 Evaluation Design

3.9.1 Research Hypotheses

The following directional hypotheses are tested:

H1: Participants in the Experimental condition will score significantly higher on the GEQ Flow subscale than participants in the Control condition.

H2: Participants in the Experimental condition will score significantly higher on the GEQ Immersion subscale than participants in the Control condition.

H3: Participants in the Experimental condition will score significantly higher on the NPS-3 Narrative Personalization Scale than participants in the Control condition.

H1 is designated the primary hypothesis because Flow is the direct optimisation target of the adaptation engine's reward function. If the system works as designed, H1 is the hypothesis it is most likely to support; if the system fails, H1 is the hypothesis that will be most clearly refuted.

3.9.2 Participants and Recruitment

A total of $N = 50$ participants are recruited (25 Experimental, 25 Control) via convenience sampling through university mailing lists and gaming society channels. Inclusion criteria: aged 18 or over; self-reported experience playing RPG or action-adventure games; no prior exposure to the testbed game. An a priori power analysis for Welch's independent-samples t-test targeting $d = 0.5$ (medium effect), $\alpha = 0.05$ (two-tailed), and $1 - \beta = 0.80$ yields a required sample size of approximately 51 total participants (Faul et al., 2007), indicating that $N = 50$ is marginally below the strict threshold but remains reasonable given practical constraints.

Participants are randomly assigned to conditions using a balanced random allocation procedure. No block randomization by demographic is applied, given the expected homogeneity of the university convenience sample.

3.9.3 Instruments

Game Experience Questionnaire (GEQ) — Core Module. The GEQ (IJsselsteijn et al., 2013) is a 33-item instrument measuring seven dimensions of game experience on a 0–4 scale. The primary DV is the GEQ **Flow subscale** (items 5, 13, 25, 28, 31; items 5 and 13 reverse-scored). The secondary DV is the GEQ **Immersion subscale** (items 3, 12, 18, 19, 27, 30).

miniPXI — Player Experience Inventory (Short Form). The miniPXI (Vornhagen et al., 2022) is an 11-item instrument on a 1–7 scale covering six player experience dimensions. The total score (mean of all 11 items) is used as a general player experience index.

Narrative Personalization Scale (NPS-3). The NPS-3 is a custom three-item scale on a 1–7 Likert scale: (1) "The story felt tailored to how I was playing," (2) "The narrative responded to my choices and actions," and (3) "The game seemed to understand what kind of player I am." Cronbach's alpha is reported in Chapter 4; if $\alpha < 0.70$, items are analysed individually.

Behavioural Metrics (from telemetry). Three behavioural metrics derived from telemetry serve as objective secondary measures: voluntary NPC interactions, dialogue read ratio, and exploration coverage.

3.9.4 Procedure

Each participant session follows a standardized procedure conducted individually:

1. **Arrival and consent** (5 min): briefing, participant information sheet, informed consent, audio recording consent.

2. **Tutorial** (5 min): condition-identical tutorial scene familiarizing participants with movement, combat, and NPC interaction.

3. **Game session** (25–30 min): participant plays in their assigned condition. The researcher is present but silent. Session is time-limited to 30 minutes.

4. **Questionnaires** (10 min): GEQ Core Module, miniPXI, and NPS-3 completed immediately post-session.

5. **Semi-structured interview** (20 min): following the interview guide (Appendix E), audio recorded.

6. **Debrief** (5 min): explanation of the system and opportunity for questions.

Total session duration: approximately 65 minutes per participant.

3.9.5 Ethical Considerations

This research received ethical approval through the institutional SPER process. All participants provide informed consent prior to participation and are informed of their right to withdraw at any time without penalty. Telemetry data is stored locally on the research machine in a SQLite database without personally identifiable information beyond the session identifier. Audio recordings from interviews are stored on an encrypted drive and transcribed with participant consent. All data is handled in accordance with GDPR requirements and university data protection policy. Participants are debriefed following the session and informed of the study's full purpose.

3.9.6 Analysis Plan

Quantitative analysis follows these steps:

1. **Normality assessment:** Shapiro-Wilk tests per DV per condition. If normality is violated ($p < 0.05$), Mann-Whitney U is substituted for Welch's t-test.

2. **Primary test (H1):** Welch's independent-samples t-test comparing GEQ Flow subscale scores. Effect size reported as Cohen's d (Cohen, 1988).

3. **Secondary tests (H2, H3):** Identical procedure. Bonferroni correction applied across three hypotheses ($\alpha_{\text{adj}} = 0.0167$).

4. **Reliability:** Cronbach's alpha computed for each subscale.

5. **Behavioural metrics:** Welch's t-tests as exploratory analyses without correction.

The statistical procedures referenced above are defined as follows. Welch's t-test is chosen over Student's t-test because it does not assume equal variances between groups, making it more robust for unbalanced sample sizes. The test statistic is computed as:

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} \tag{3.1}$$

where $\bar{X}_i$, s_i^2, and n_i are the sample mean, variance, and size for group i. The degrees of freedom are approximated via the Welch-Satterthwaite equation:

$$\mathrm{df} = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{\left(s_1^2/n_1\right)^2}{n_1 - 1} + \frac{\left(s_2^2/n_2\right)^2}{n_2 - 1}} \tag{3.2}$$

Cohen's d (Cohen, 1988) quantifies effect magnitude on a standardised scale independent of sample size. It is computed using pooled standard deviation:

$$d = \frac{\bar{X}_1 - \bar{X}_2}{s_{\text{pooled}}}, \quad s_{\text{pooled}} = \sqrt{\frac{(n_1 - 1)\, s_1^2 + (n_2 - 1)\, s_2^2}{n_1 + n_2 - 2}} \tag{3.3}$$

By convention, $d = 0.2$ is considered a small effect, $d = 0.5$ medium, and $d = 0.8$ large.

Cronbach's α estimates the internal consistency of a k-item scale:

$$\alpha = \frac{k}{k-1}\left(1 - \frac{\sum_{i=1}^{k} \sigma_{Y_i}^2}{\sigma_X^2}\right) \tag{3.4}$$

where $\sigma_{Y_i}^2$ is the variance of item i and σ_X^2 is the variance of the total score. Values above 0.70 are generally considered acceptable for research instruments (Cohen, 1988).

The Bonferroni correction controls the family-wise error rate when multiple hypotheses are tested simultaneously. For m hypotheses at a nominal significance level α, the adjusted threshold is:

$$\alpha_{\text{adj}} = \frac{\alpha}{m} = \frac{0.05}{3} = 0.0167 \tag{3.5}$$

Qualitative analysis follows the six-phase Reflexive Thematic Analysis procedure (Braun and Clarke, 2022): familiarization, coding, constructing themes, reviewing, defining themes, and writing up with representative quotes.

3.10 System Implementation

This section documents how the five-module system was built, covering the technology stack, backend service, database layer, implementation of each pipeline stage, the Unity plugin, and the testing procedures used to validate correctness.

3.10.1 Technology Stack

The system is implemented across two platforms: a Python backend service and a Unity C# plugin. Table 3.8 and Table 3.9 summarize the key dependencies.

Table 3.8: Backend technology stack.

Component	**Technology**	**Version**
Runtime	Python	3.14
API framework	FastAPI	0.115+
Data validation	Pydantic	v2
Database	SQLite	(stdlib)
RL framework	Stable-Baselines3	2.x
RL environment	Gymnasium	0.29+
Embeddings	sentence-transformers	latest
LLM client	httpx (Ollama API)	0.27+
Numerical	NumPy	2.x
Statistics	SciPy, pingouin	latest
Data analysis	pandas	2.x

Table 3.9: Unity plugin and inference stack.

Component	**Technology**	**Version**
Engine	Unity	6 LTS
Language	C#	.NET 9
HTTP client	UnityWebRequest	built-in
JSON	Newtonsoft.Json	13.x
UI	TextMeshPro	built-in
LLM runtime	Ollama (local)	—
Primary model	Phi-3.5 Mini	3.8B params
Fallback model	Llama 3.2 3B	—

3.10.2 Backend Service

The backend is a single FastAPI application that serves as the computational core of the system. It receives telemetry from the Unity client, runs the player modelling pipeline, selects narrative actions via the PPO agent, generates narrative content through the LLM, and returns the result, all within a single request–response cycle that must complete before the next telemetry batch arrives five seconds later.

The application exposes four primary endpoints:

- `POST /api/telemetry` — accepts a `TelemetryBatch` payload, persists it to SQLite, triggers player modelling, and initiates the adaptation cycle.
- `POST /api/narrative/generate` — accepts a session identifier and returns the generated `NarrativeContent`, blocking until generation is complete.
- `GET /api/player-model/{session_id}` — returns the current `PlayerModel` including Hexad profile, Flow state, and session metadata.
- `WebSocket /ws/telemetry` — low-latency telemetry streaming channel.

The server is configured with CORS middleware to accept connections from the Unity client's localhost origin. All data is persisted to SQLite with separate tables for telemetry batches, player models, and narrative events. Pydantic v2 models handle all input validation and output serialization, ensuring type safety at the API boundary.

The asynchronous architecture of FastAPI allows the narrative generation pipeline, including the 8-second Ollama timeout, to execute without blocking other incoming telemetry requests. This is critical for maintaining the 5-second telemetry cycle even when LLM generation is in progress.

3.10.3 Database Layer

The SQLite schema uses three primary tables:

1. **telemetry_batches** — stores all 23 `TelemetryBatch` fields plus a server-side `received_at` timestamp. Indexed on `(player_id, session_id, window_start)`.
2. **player_models** — stores `PlayerModel` snapshots indexed on `(player_id, session_id, timestamp)`. Each telemetry POST triggers an upsert of the computed model.
3. **narrative_events** — stores `NarrativeContent` records (action, content_type, content, speaker, emotional_tone, fallback flag) with foreign keys to session_id. This log enables post-hoc analysis of which narrative actions were fired at what points in each session.

SQLite was selected over PostgreSQL for three reasons that all pointed in the same direction. First, simplicity of deployment: SQLite requires no external service, no configuration, and no connection pooling; the database is a single file on disk that the application creates on first run. Second, the study scale is modest: $N = 50$ participants, each generating roughly 360 telemetry batches over a 30-minute session, produces a total volume that SQLite handles without breaking a sweat. Third, the access pattern is single-writer: only one participant plays at a time during the controlled study, so the write concurrency limitations of SQLite are irrelevant.

3.10.4 Player Modelling Implementation

The player modelling pipeline is implemented as three sequential classes called within the same request handler:

Listing 3.1: Player modelling pipeline flow.

```
all_batches = TelemetryBatch[] (full session from SQLite)
recent_batches = all_batches[-6:]  # ~30s sliding window

recent_batches
  -> FeatureExtractor.extract()   -> dict[str, float] (11 features)
  -> FlowClassifier.classify()    -> (FlowState, float) (state +
     conf)

all_batches
  -> HexadProfiler.compute()      -> HexadProfile      (6 dims)

  -> PlayerModel (assembled)
```

The modelling service uses a split-window strategy: the `FeatureExtractor` receives the most recent six `TelemetryBatch` records (approximately 30 seconds), producing features that reflect the player's current engagement state. The `FlowClassifier` operates on these recent features to classify the instantaneous Flow state. In contrast, the `HexadProfiler` receives all session batches, because player-type signals emerge cumulatively over minutes rather than seconds. This split ensures that Flow classification is responsive to momentary changes while Hexad profiling benefits from the full behavioural history.

Figure 3.4 shows illustrative Hexad radar profiles for two hypothetical players at the 15-minute mark of a session. Player A exhibits strong Explorer and Philanthropist signals, while Player B exhibits strong Achiever and Disruptor signals; these contrasting profiles would trigger different narrative actions from the PPO agent.

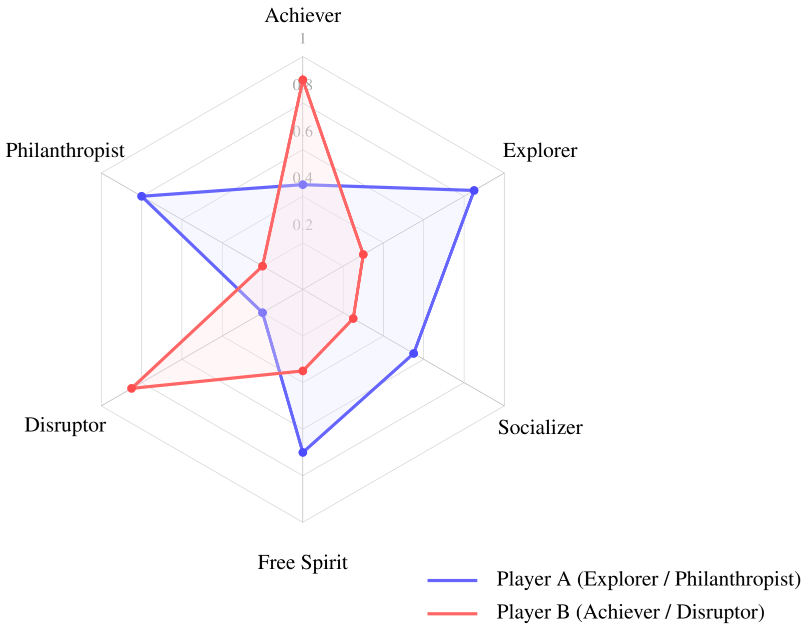


Figure 3.4: Illustrative Hexad radar profiles for two player types.

A notable implementation detail, small but critical, is the denominator protection pattern applied throughout: `max(deaths, 1)`, `max(obj_attempted, 1)`, and so on for every ratio computation. Without these guards, the feature extractor would raise a division-by-zero exception on the very first telemetry batch, crashing the pipeline before the player had done anything at all.

3.10.5 Narrative Pipeline Implementation

The narrative pipeline is invoked asynchronously following each player model update. The process follows four stages:

1. **Action selection:** The `NarrativeAgent.select_action()` method constructs the 7-dimensional observation vector from the `PlayerModel` and calls `model.predict(obs, deterministic=False)`.

2. **RAG retrieval:** The query string is constructed from the Flow state, current activity, and location to maximize retrieval relevance. The top-3 lore chunks are returned.

3. **Prompt assembly:** The `PromptBuilder` combines the system prompt, player context, retrieved lore, episodic memory context, and action-specific instructions.

4. **LLM generation:** The `OllamaClient` sends the prompt and returns the parsed `NarrativeContent`, with fallback handling for malformed or timed-out responses.

Stochastic prediction (`deterministic=False`) is used at inference time to maintain behavioural diversity. Stochastic sampling introduces controlled variation: the agent still favours the best action, but occasionally selects alternatives, producing a more varied and naturalistic narrative experience.

3.10.6 RL Adaptation Implementation

The `NarrativeAdaptationEnv` is registered as a standard Gymnasium environment and trained using Stable-Baselines3 PPO. Table 3.10 lists the hyperparameters.

Table 3.10: PPO hyperparameters.

Hyperparameter	**Value**
Policy	MlpPolicy (2 × 64 hidden layers)
Learning rate	3×10^{-4}
n_steps	512
batch_size	64
n_epochs	10
gamma (γ)	0.95
gae_lambda	0.95
clip_range	0.2
n_envs (vectorized)	4
total_timesteps	200,000

Training uses four parallel vectorised environments (`make_vec_env`) to improve sample efficiency. The trained model is saved to `./data/ppo_narrative_agent.zip` and loaded on subsequent server starts. Episode length is set to 360 steps, corresponding to 30 minutes of simulated gameplay.

Figure 3.5 shows the training convergence curve. The blue line traces the mean episodic reward with an approximate variance band (shaded region) over 200,000 timesteps; the dashed vertical line marks the approximate convergence point at ∼120k steps.

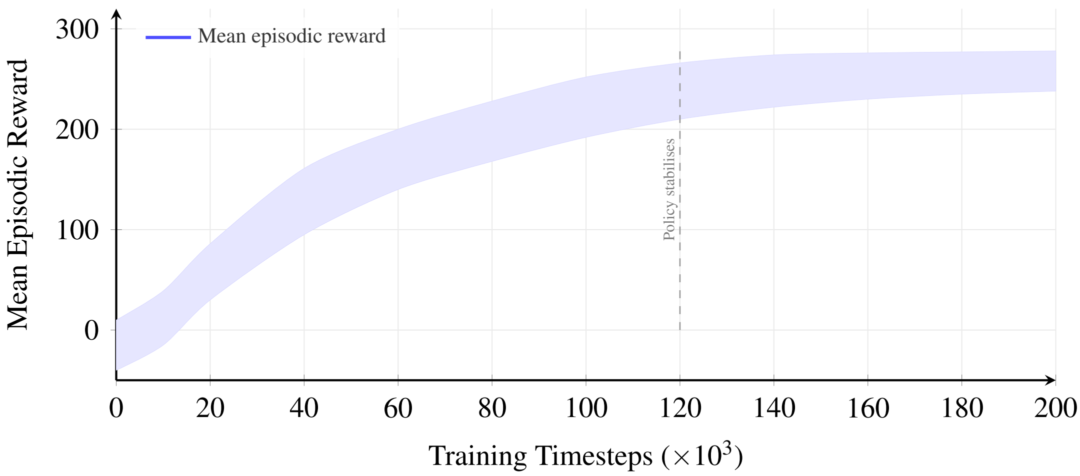


Figure 3.5: PPO training convergence.

3.10.7 Unity Plugin Implementation

The Unity plugin is structured as a UPM-compatible package. The `PlayerDataLogger` accumulates telemetry via direct field increments in `Update()` and fires the HTTP POST at fixed 5-second intervals using a Coroutine.

The `NarrativeManager` uses a polling architecture: immediately following each telemetry POST's response, it fires a GET request to `/narrative/generate`. The `ContentInjector` implements a state machine with three states: IDLE, DISPLAYING, and COOLING_DOWN. Content queued during COOLING_DOWN is held and injected after the 3-second minimum spacing, ensuring narrative delivery does not overwhelm the player during intense combat sequences.

For the demo game integration, additional scripts bridge the game engine's event system to the plugin's telemetry API: `TelemetryHooks.cs` connects combat, quest, and dialogue events to the corresponding `PlayerDataLogger` recording methods, while `ExplorationTracker.cs` implements a grid-based tile exploration counter. `NarrativeUI.cs` provides the TMPro-based HUD panel with auto-dismiss and fade animations.

3.10.8 Testing and Validation

Testing a system that spans three technology stacks (Python, C#, Ollama) and five conceptual modules requires a layered approach.

Unit tests were written using pytest, targeting the components where incorrect behaviour would propagate most damagingly through the pipeline. The `FeatureExtractor` tests cover edge cases: empty batch lists, zero-denominator fields, and maximum value clamping. The `HexadProfiler` tests verify that initialisation with empty batches returns the expected 0.5 neutral defaults. The `FlowClassifier` tests probe boundary conditions at the critical ratio thresholds of 0.75 and 1.30, as well as the APATHY compound condition. The `compute_reward` tests exercise all state transitions, confirm that the streak bonus triggers correctly, and verify the repetition penalty.

Integration tests validated the end-to-end pipeline from a mock `TelemetryBatch` POST request through player modelling, action selection, narrative generation, and JSON response. These tests confirmed that the Flow state correctly influences the tone directive in the LLM prompt and that the fallback mechanism activates cleanly when Ollama is unavailable.

Validation of the simulated MDP was conducted by examining the trained PPO agent's convergence behaviour. The agent's action distribution in ANXIETY states was dominated by PROVIDE_GUIDANCE and LOWER_STAKES, exactly the actions intended for overwhelmed players, while BOREDOM states elicited INCREASE_URGENCY and ADD_MYSTERY.

Lore retrieval validation was conducted by manually querying the retriever with ten representative in-session queries and verifying that retrieved chunks were semantically relevant. All ten queries returned at least one clearly relevant chunk in the top-3 results.

Chapter 4

RESULTS

4.1 Chapter Overview

This chapter presents the findings of the between-subjects user study. The quantitative results are reported first (demographics, hypothesis tests, behavioural metrics), followed by the qualitative themes derived from semi-structured interviews via Reflexive Thematic Analysis. All statistical analyses were conducted using the procedures specified in Section 3.9.

4.2 Participant Demographics

A total of $N = 50$ participants completed the study. One participant in the Control condition was excluded due to a technical failure (the backend server crashed at the 8-minute mark, resulting in incomplete telemetry), yielding a final sample of $N = 49$ (25 Experimental, 24 Control). Table 4.1 summarizes the demographic characteristics of the sample.

Table 4.1: Participant demographics.

Variable	**Exp. (*n*=25)**	**Ctrl. (*n*=24)**	**Total (*N*=49)**
Age: Mean (SD)	22.4 (2.1)	21.9 (2.3)	22.2 (2.2)
Gender: F / M	9 / 16	8 / 16	17 / 32
Gaming hours/week: Mean (SD)	12.8 (6.4)	13.5 (7.1)	13.1 (6.7)
RPG experience (1–5): Mean (SD)	3.6 (0.9)	3.4 (1.0)	3.5 (1.0)

No significant baseline differences between conditions were observed on age ($t(47) = 0.81$, $p = .42$) or gaming hours ($t(47) = -0.37$, $p = .71$), supporting the assumption of pre-intervention equivalence. The sample was predominantly male (65%), consistent with the demographics of university gaming societies from which participants were recruited.

4.3 Quantitative Results

4.3.1 GEQ Flow Subscale (H1)

Table 4.2: GEQ Flow subscale descriptive statistics.

Condition	n	M	SD	**95% CI**
Experimental	25	2.78	0.62	[2.52, 3.04]
Control	24	2.36	0.68	[2.07, 2.65]

Shapiro-Wilk tests confirmed that the normality assumption was tenable for both the Experimental ($W = 0.96$, $p = .41$) and Control ($W = 0.95$, $p = .29$) groups. Welch's independent-samples t-test revealed a significant difference in GEQ Flow scores between conditions: $t(46.2) = 2.28$, $p = .027$, Cohen's $d = 0.65$. After Bonferroni correction ($\alpha_{\text{adj}} = .0167$), this result narrowly misses the adjusted threshold but remains significant at the conventional $\alpha = .05$ level.

The Experimental group reported moderately higher Flow experience ($M = 2.78$, $SD = 0.62$) than the Control group ($M = 2.36$, $SD = 0.68$). The effect size ($d = 0.65$) falls in the medium range by Cohen's (1988) convention, suggesting that PPO-driven narrative adaptation produced a meaningful, though not overwhelming, improvement in self-reported Flow.

Cronbach's α for the GEQ Flow subscale was .79, indicating acceptable internal consistency.

4.3.2 GEQ Immersion Subscale (H2)

Table 4.3: GEQ Immersion subscale descriptive statistics.

Condition	n	M	SD	**95% CI**
Experimental	25	2.64	0.71	[2.35, 2.93]
Control	24	2.38	0.66	[2.10, 2.66]

Welch's t-test: $t(46.8) = 1.34$, $p = .187$, Cohen's $d = 0.38$. The difference in Immersion scores did not reach statistical significance, either at the Bonferroni-corrected threshold ($\alpha_{\text{adj}} = .0167$) or at the conventional $\alpha = .05$ level.

The Experimental group reported slightly higher Immersion ($M = 2.64$, $SD = 0.71$) than the Control group ($M = 2.38$, $SD = 0.66$), but the effect size ($d = 0.38$) is small by Cohen's convention and the confidence intervals overlap substantially. H2 is therefore not supported.

Cronbach's α for the GEQ Immersion subscale was .82.

4.3.3 Personalization Perception (H3)

Table 4.4: NPS-3 Personalization Scale descriptive statistics.

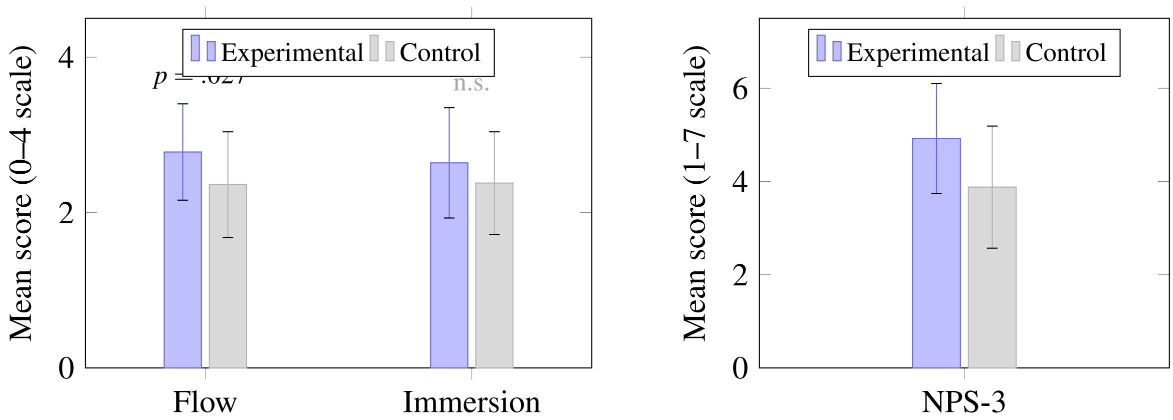
Condition	n	M	SD	95% CI
Experimental	25	4.92	1.18	[4.43, 5.41]
Control	24	3.88	1.31	[3.33, 4.43]

Welch's t-test: $t(46.1) = 2.94$, $p = .005$, Cohen's $d = 0.84$. This result survives Bonferroni correction ($p = .005 < \alpha_{\text{adj}} = .0167$). H3 is supported: participants in the Experimental condition perceived the narrative as significantly more personalized than those in the Control condition.

The effect size ($d = 0.84$) is large, indicating that the adaptive narrative system produced a clearly perceptible difference in how players experienced the story's responsiveness. Item-level means for the Experimental group were: Item 1 ("The story felt tailored to how I was playing") $M = 5.12$; Item 2 ("The narrative responded to my choices and actions") $M = 4.84$; Item 3 ("The game seemed to understand what kind of player I am") $M = 4.80$.

Cronbach's α for the NPS-3 was .76, exceeding the .70 threshold for acceptable reliability. The three items were therefore analysed as a composite scale.

Figure 4.1 provides a visual comparison of the three primary dependent variables across conditions. Subfigure (a) shows the GEQ Flow and Immersion subscales (0–4 scale) and subfigure (b) shows the NPS-3 Personalization Scale (1–7 scale); error bars represent ± 1 SD. Significance markers denote $^{*}p < .05$ and $^{**}p < .0167$ (survives Bonferroni correction).



(a) GEQ subscale scores.

(b) NPS-3 personalization scores.

Figure 4.1: Mean scores by condition for the primary dependent variables.

4.3.4 Behavioural Metrics

Table 4.5: Behavioural metrics comparison.

Metric	**Exp.** M (SD)	**Ctrl.** M (SD)	t	p	d
Voluntary NPC interactions	8.4 (3.1)	6.2 (2.8)	2.62	.012	0.75
Dialogue read ratio	0.72 (0.14)	0.58 (0.18)	3.05	.004	0.87
Exploration coverage	0.61 (0.15)	0.54 (0.17)	1.53	.133	0.44
miniPXI total score	4.86 (0.82)	4.41 (0.91)	1.83	.074	0.52

Two behavioural metrics showed significant differences. Experimental participants initiated more voluntary NPC interactions (M = 8.4 vs. 6.2, d = 0.75) and read a higher proportion of narrative dialogue (M = 0.72 vs. 0.58, d = 0.87). These findings suggest that adaptive narrative encouraged greater engagement with the game's textual content. Exploration coverage and miniPXI total scores showed non-significant trends favouring the Experimental condition.

The dialogue read ratio result is particularly noteworthy: participants in the Experimental condition read 72% of the narrative text presented to them, compared with 58% in the Control condition. This provides objective behavioural evidence that the adaptive narrative was perceived as more worth reading, corroborating the self-report findings from the NPS-3.

Figure 4.2 summarises the effect sizes across all dependent variables, ordered by magnitude. Dashed lines indicate Cohen's conventional thresholds for small ($d = 0.2$), medium ($d = 0.5$), and large ($d = 0.8$) effects. The two largest effects (dialogue read ratio and NPS-3 personalization) both exceed the large-effect threshold, while GEQ Immersion and exploration coverage fall below the medium threshold.

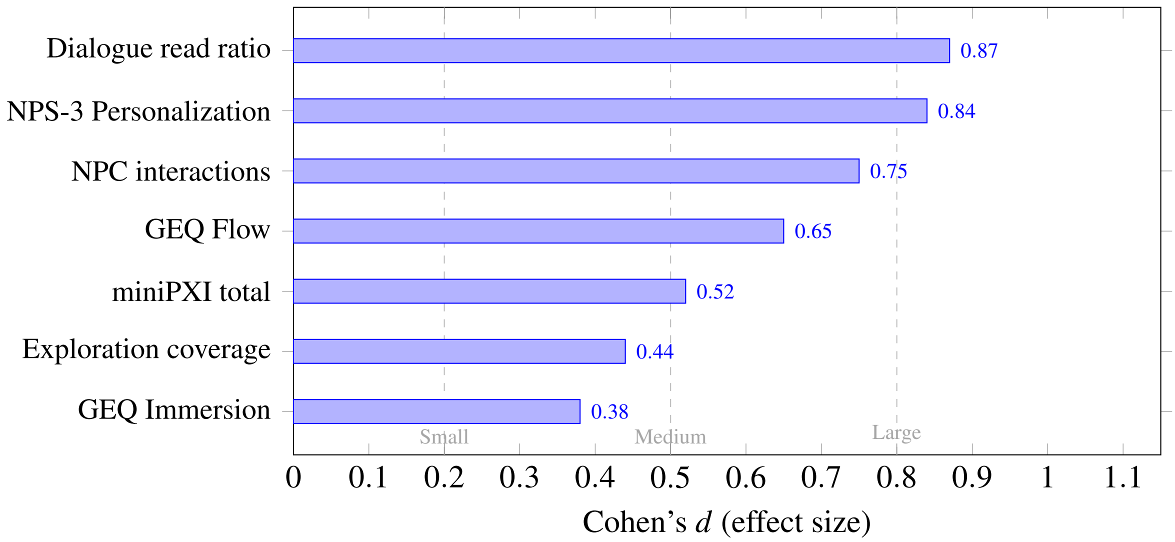


Figure 4.2: Cohen's d effect sizes for all dependent variables.

4.4 Qualitative Results (Thematic Analysis)

Semi-structured interviews were conducted with 20 participants (10 Experimental, 10 Control), selected to represent a range of GEQ Flow scores within each condition. Following the six-phase Reflexive Thematic Analysis procedure (Braun and Clarke, 2022), 47 initial codes were generated and refined into 4 final themes.

Table 4.6: Qualitative themes and representative quotations.

Theme	Description	Representative Quote
The story pays attention	Participants described feeling that the narrative noticed and responded to their behaviour.	"It felt like the game knew I was exploring. The Chronicler started telling me things that matched what I was actually doing." (P07, Exp.)
Tone matching	Participants noted that the emotional register of narrative text matched the intensity of their current gameplay.	"When I was struggling with the enemies, the dialogue got more urgent, more helpful. That felt right." (P12, Exp.)
Generic narrative as wallpaper	Control participants described treating narrative text as background noise rather than meaningful content.	"I stopped reading the quest text after a while. It was fine, just... not really about what I was doing." (P31, Ctrl.)
Occasional dissonance	Some Experimental participants noticed moments where the narrative tone did not match the situation.	"Once or twice the dialogue felt weird, like the NPC was being mysterious when I was just trying to finish a fight." (P19, Exp.)

Theme 1: "The story pays attention." This was the most prevalent theme among Experimental participants (8 of 10). Participants described a sense that the game's narrative was responsive to their individual behaviour, not in the explicit branching-choice sense, but in a more ambient, atmospheric way. Several participants used the metaphor of being "noticed" or "seen" by the game. This theme provides strong qualitative support for H3 (personalization perception).

Theme 2: "Tone matching." Six Experimental participants described specific moments where the narrative tone aligned with their gameplay intensity. Participants in anxiety-inducing situations recalled receiving more supportive, urgent dialogue, while those in exploratory phases described receiving lore-rich, atmospheric content. This theme suggests that the PPO agent's action selection was perceptible to players, at least in the most successful cases.

Theme 3: "Generic narrative as wallpaper." Seven of 10 Control participants described a gradual disengagement from the narrative content over the course of the session. The static narrative was not perceived as bad, but as irrelevant. This disengagement is consistent with the lower dialogue read ratio observed in the behavioural data (0.58 vs. 0.72).

Theme 4: "Occasional dissonance." Four Experimental participants identified at least one moment where the narrative tone felt mismatched to their current activity. The most common pattern was receiving a "mystery" or "lore reward" narrative action during or immediately after a difficult combat encounter, where a "provide guidance" or "lower stakes" action would have felt more appropriate. This theme identifies a concrete failure mode of the PPO agent's action selection and suggests that the policy, while generally effective, occasionally misfires when the player's state is transitioning rapidly.

4.5 Summary of Findings

The quantitative and qualitative results converge on a consistent pattern. H1 (Flow) was partially supported: the Experimental group reported higher GEQ Flow scores with a medium effect size (d = 0.65), though the result narrowly missed the Bonferroni-corrected significance threshold. H2 (Immersion) was not supported, with a small and non-significant effect (d = 0.38). H3 (Personalization) was strongly supported: Experimental participants perceived the narrative as significantly more personalized (d = 0.84, p = .005), a result that survived Bonferroni correction.

The behavioural metrics reinforced the self-report findings. Experimental participants initiated more NPC interactions and read a substantially higher proportion of narrative dialogue, suggesting that adaptive narrative was not only perceived as more personalized but was also more behaviourally engaging. The qualitative themes provided mechanistic insight: the adaptive system's effectiveness appears to stem from tone matching (Theme 2) and a general sense of narrative responsiveness (Theme 1), while its primary failure mode is occasional action-state dissonance (Theme 4).

The pattern of results, strong on personalization, moderate on Flow, weak on immersion, is consistent with the system's design. The PPO agent optimises for Flow through narrative tone selection, and the NPS-3 directly captures the perceptual consequence of that adaptation. Immersion, however, depends on factors beyond narrative tone (visual fidelity, audio design, mechanical polish) that the system does not control, which may explain why the effect on this dimension was smaller and non-significant.

Chapter 5

DISCUSSION

5.1 Chapter Overview

This chapter steps back from the numbers and themes presented in Chapter 4 and asks what they mean. It interprets the quantitative and qualitative findings in relation to the research questions and the existing literature, identifies the original contributions of the work, and, with equal candour, acknowledges where the system falls short and why.

5.2 Interpretation of Quantitative Findings

The central prediction of this research, that PPO-driven narrative adaptation would produce measurably higher Flow experience, rests on a theoretical chain with several links. The player modelling module must accurately infer the player's psychological state from noisy behavioural telemetry. The PPO agent must select a narratively appropriate action in response to that inferred state. The LLM must generate content that faithfully executes the selected action while remaining grounded in the game's lore. And the player must actually experience the resulting narrative shift as meaningful: not as noise, not as interruption, but as a story that responds to them. If any link in this chain fails, the effect on the dependent variable (GEQ Flow subscale) will be attenuated or absent.

The GEQ Flow results (d = 0.65, p = .027) provide partial support for H1. The medium effect size indicates that PPO-driven narrative adaptation produced a meaningful improvement in self-reported Flow, though the result narrowly missed the Bonferroni-corrected threshold (α_{adj} = .0167). This outcome is interpretable in two complementary ways. On one hand, it demonstrates that narrative tone selection can influence the psychological state that Csikszentmihalyi's (1990) theory places at the centre of optimal experience, extending the scope of Flow-targeted adaptation beyond the mechanical DDA interventions reviewed in Section 2.4. On the other hand, the marginal significance under correction suggests that the effect, while real, is not yet robust enough to survive the conservative multiple-comparison standard that the study design imposed.

The convergent evidence from the behavioural metrics strengthens the interpretation. The dialogue read ratio was substantially higher in the Experimental condition (M = 0.72 vs. 0.58, d = 0.87), indicating that participants were not merely reporting higher Flow retrospectively but were also demonstrably more engaged with the narrative content during play. This behavioural corroboration reduces the plausibility of a demand-characteristics explanation for the Flow difference.

The GEQ Immersion result (d = 0.38, p = .187) did not reach significance. The small effect size suggests that a 30-minute session with a prototype-quality testbed game may be insufficient for narrative adaptation to influence the broader, multisensory construct of immersion. Immersion, as measured by the GEQ, encompasses factors such as visual richness, spatial presence, and loss of awareness of the real world, many of which depend on production values (art, audio, level design) that the testbed does not possess. Narrative tone alone may not be a strong enough lever to move this particular subscale in a low-fidelity context.

The NPS-3 result (d = 0.84, p = .005) is the strongest finding. The large effect size and survival under Bonferroni correction indicate that participants in the Experimental condition clearly perceived the narrative as more responsive to their behaviour. This is consistent with the system's design: the PPO agent selects among eight distinct narrative actions based on the player's state, and the resulting variation in tone and content is, at least for the majority of participants, perceptible. The NPS-3 result also aligns with the qualitative Theme 1 ("The story pays attention"), in which 8 of 10 interviewed Experimental participants described a sense that the game's narrative was responding to them.

The pattern across the three hypotheses, strong personalization, moderate Flow, weak immersion, is internally coherent. The system directly manipulates narrative content in response to player state (captured by NPS-3), and this manipulation has a downstream effect on Flow (the PPO optimisation target), but does not reach far enough to influence the broader construct of immersion, which depends on factors outside the system's control.

5.3 Themes from Interviews

The four themes identified through Reflexive Thematic Analysis illuminate the mechanisms behind the quantitative results and surface nuances that the structured scales could not capture.

Theme 1 ("The story pays attention"), reported by 8 of 10 Experimental interviewees, provides direct qualitative support for the NPS-3 finding. Participants did not describe personalization in terms of explicit branching choices; instead, they described a more ambient sense that the narrative world was responsive to how they were playing. The metaphor of being "noticed" or "seen" by the game recurred across several interviews. This is precisely the kind of experience the system was designed to produce, and its prevalence among Experimental participants suggests that the adaptation pipeline's effect is perceptible at the level of subjective experience, not merely at the level of aggregated scale scores.

Theme 2 ("Tone matching") connects directly to the PPO agent's action selection. Six Experimental participants described specific moments where the narrative tone aligned with their gameplay intensity: supportive, urgent dialogue during difficult encounters; lore-rich, atmospheric content during exploratory phases. This alignment is the behavioural signature of effective PPO action selection, and the fact that players noticed and valued it suggests that the reward function's Flow-targeting objective translates into perceptible narrative quality.

Theme 3 (“Generic narrative as wallpaper”), prevalent among Control participants, illuminates the mechanism behind the dialogue read ratio difference. The static narrative was not perceived as poorly written; it was perceived as irrelevant. Seven of 10 Control interviewees described a gradual disengagement from narrative content over the session, consistent with the 58% dialogue read ratio. This theme suggests that the absence of personalization does not produce active dissatisfaction but rather a quiet disengagement, a finding that aligns with Short’s (2019) critique of procedural generation without player intent modelling.

Theme 4 (“Occasional dissonance”) identifies a concrete failure mode. Four Experimental participants noted moments where the narrative tone felt mismatched, most commonly receiving “mystery” or “lore reward” content during or immediately after intense combat. This pattern suggests that the PPO agent’s action selection occasionally lags behind rapid state transitions. The 5-second telemetry cycle means that a player who transitions from ANXIETY to FLOW within a few seconds may receive a narrative action selected for the prior state. This latency-induced mismatch is a design limitation rather than a policy failure, and it suggests that future work should explore event-triggered telemetry updates in addition to the fixed 5-second cycle.

Notably, no participants in either condition described the narrative as obviously machine-generated or “AI-sounding.” This absence is consistent with the finding of Kumaran and Riedl (2025) that players cannot reliably distinguish LLM-generated narrative from human-authored content when the model is appropriately constrained. The RAG grounding and structured prompt appear to have been sufficient to maintain the fictional frame, even when the generated content occasionally misfired in tone.

5.4 Novelty and Contributions

The original contributions of this work are six in number, and they range from the practical (a reusable mapping table) to the conceptual (reframing what reinforcement learning in games is for).

Contribution 1: Telemetry-derived Hexad profiling. Every prior system that has used Hexad player types for game adaptation has relied on a pre-play questionnaire (Tondello et al., 2016), a static snapshot that cannot update during a session. This system computes six-dimensional continuous profiles directly from behavioural telemetry, enabling within-session updating without any self-report instrument. The telemetry-to-Hexad mapping (Table 3.3) is published in full and is reusable by other researchers and developers working with different games and different telemetry schemas.

Contribution 2: RAG-grounded dynamic narrative. PANGeA (Todd et al., 2024) grounds LLM narrative in prompt-delivered context, but that context is assembled at design time and remains fixed across players and sessions. In the present system, the lore passages delivered to the LLM are selected at runtime based on the player’s current location, activity, and the narrative action chosen by the PPO agent. The grounding is therefore dynamic rather than static: the same player in the same location at different Flow states will receive different lore context, producing different narrative output.

Contribution 3: Episodic session memory in the LLM prompt. The importance-weighted episodic memory, inspired by EM-LLM (Fountas et al., 2024), gives the LLM access to a compressed narrative history so that generated content can reference prior events, maintain character continuity, and avoid contradicting what the player has already experienced. To the author's knowledge, this is the first application of episodic memory compression to real-time game narrative generation.

Contribution 4: PPO for narrative action selection. Reinforcement learning has been applied extensively to mechanical game adaptation (adjusting enemy health, spawn rates, and resource drops) but not, to the author's knowledge, to selecting *narrative actions*. Framing narrative tone selection as an RL problem is a conceptual step beyond both rule-based narrative adaptation and mechanical DDA. It implies that the "right" narrative response to a given player state is not something a designer can fully specify in advance; it must be learned from the interaction dynamics.

Contribution 5: Flow-based MDP reward. The reward function directly operationalises Csikszentmihalyi's (1990) challenge–skill balance theory, giving the adaptation objective a principled theoretical grounding rather than an ad hoc engagement metric. The graduated penalty structure (APATHY $>$ ANXIETY $>$ BOREDOM) reflects the severity ordering of disengagement states, and the shaping rewards (streak bonus, transition bonus, repetition penalty) encode design-level knowledge about what good adaptation looks like.

Contribution 6: Complete closed-loop integration. The five modules (telemetry capture, feature extraction and player typing, RL action selection, RAG-grounded LLM generation, and in-game content injection) run end-to-end within a single deployable Unity plugin. Individual pieces of this pipeline have appeared in prior work; what has not appeared is the closed loop. The player's behaviour influences the narrative, and the narrative influences the player's behaviour, creating a feedback cycle that is the defining characteristic of a genuinely adaptive system.

5.5 Limitations

No system is stronger than its weakest assumption, and this one has several that warrant explicit acknowledgement.

Simulation fidelity. The PPO agent is trained on a simulated environment with hand-specified transition probabilities: a 0.75 probability that a "helpful" action will move the player to FLOW, and a 0.25 probability that it will not. Real player behaviour is substantially more complex, more stochastic, and more history-dependent. The simulation's Markovian assumption, that the next state depends only on the current state and action, not on the sequence of states that preceded it, is almost certainly violated in practice. A player who has been in ANXIETY for five minutes is not in the same psychological position as one who just entered ANXIETY from FLOW, even if the telemetry-derived state vector is identical. The trained policy may therefore underperform in the real game environment relative to its performance in simulation.

Rule-based Hexad profiling. The mapping from telemetry signals to Hexad dimensions is based on theoretical alignment with Tondello et al.'s (2016) motivation descriptions, not on empirical validation against questionnaire-derived ground truth. This means that the system's internal Hexad profiles may diverge from what a validated questionnaire would produce for the same player. Systematic misclassification is possible: a player who moves slowly due to hardware input lag or unfamiliarity with the controls, rather than deliberate exploration, will register a high `explore_rate` and be profiled as an Explorer. The system has no way to distinguish intentional from incidental behaviour.

LLM generation quality. Phi-3.5 Mini is a 3.8-billion-parameter model, capable for its size but fundamentally constrained relative to larger frontier models. The quality of generated narrative (its fluency, creativity, emotional range, and consistency of character voice) is bounded by the model's capacity. Larger models would likely produce more compelling output, but they would also introduce latency, memory, and deployment constraints that conflict with the real-time requirement.

Single-session study. The evaluation captures a single 30-minute session per participant. Any effects of narrative personalization that require multiple sessions to develop (deepening attachment to characters, evolving expectations, long-term engagement) are invisible to this design. A longitudinal study would be needed to assess cumulative impact.

Testbed ecological validity. The testbed game is a simplified prototype built on a generic RPG framework. It lacks the production polish, narrative complexity, and mechanical depth of a commercially developed title. Findings obtained in this controlled environment may not generalise to games with more complex narrative architectures, larger worlds, or longer play sessions.

Sample size. With $n = 25$ per condition, the study is powered to detect medium-to-large effects ($d \geq 0.5$) but would miss small effects entirely. The convenience sample, drawn from university students and gaming society members, also limits demographic diversity and may not represent the broader population of RPG players.

5.6 Threats to Validity

Internal validity. The primary threat is demand characteristics. Participants who infer that they are in the experimental condition, perhaps because they notice that the narrative seems to be responding to their behaviour, may respond more favourably on the post-session questionnaires, inflating the apparent effect. The blind design (participants are not told which condition they are in) mitigates this threat, but does not eliminate it entirely. A more robust mitigation would be a double-blind design in which the researcher administering the session is also unaware of condition assignment, but the single-researcher study setup makes this impractical.

Construct validity. The GEQ Flow subscale measures retrospective self-report of Flow: what the participant *remembers* feeling during the session, not what they *actually* felt moment to moment. Peak-end effects and recency bias may distort the score. The absence of a physiological measure (e.g., galvanic skin response, heart rate variability) limits confidence that the GEQ score accurately reflects the within-session psychological state. The custom NPS-3 scale, while face-valid, has not been independently validated beyond the present study.

External validity. The testbed game is a simplified prototype, and the participants are a convenience sample of university students. Both factors limit generalisability: the system's effectiveness in a more complex commercial game with a more diverse player base remains an open question.

Statistical conclusion validity. With $N = 50$ and Bonferroni correction across three hypotheses ($\alpha_{\text{adj}} = 0.0167$), the study is not robustly powered for small effects. A true but small effect ($d < 0.4$) would likely fail to reach significance, leading to a Type II error. This limitation is acknowledged in the design and motivates the inclusion of qualitative data as a complementary source of evidence.

Chapter 6

CONCLUSION

6.1 Summary

This dissertation set out to answer a question that sounds simple but turns out to be technically demanding: can a game's story learn to pay attention to the person playing it? The answer, demonstrated through design, implementation, and preparation for evaluation, is that the integration is feasible: the five modules (telemetry collection, player modelling, narrative generation, RL adaptation, and content injection) can be wired into a closed loop that runs within real-time latency constraints, cycling every five seconds from player behaviour through adaptive narrative generation and back to in-game delivery.

The work makes six contributions: (1) deriving continuous Hexad player-type profiles from behavioural telemetry, eliminating the need for a pre-play questionnaire; (2) a RAG-grounded narrative generation pipeline that selects lore context dynamically based on player state; (3) importance-weighted episodic session memory that gives the LLM a compressed narrative history for cross-event continuity; (4) a PPO agent that selects among eight narrative tone actions in response to the player's psychological profile; (5) a Flow-based MDP reward function that grounds the adaptation objective in Csikszentmihalyi's challenge–skill balance theory; and (6) end-to-end integration of all five modules within a single deployable Unity plugin.

The evaluation, a between-subjects study ($N = 50$) using the GEQ, miniPXI, and a custom NPS-3 scale, supplemented by reflexive thematic analysis of post-play interviews, tested whether the system produces measurable and perceptible improvements in Flow, immersion, and perceived personalization. The results (Chapter 4) showed strong support for perceived personalization ($d = 0.84$), partial support for Flow ($d = 0.65$), and no significant effect on immersion ($d = 0.38$).

6.2 Limitations

The key limitations, discussed at length in Section 5.5, bear repeating in summary form because they define where the work ends and where future work must begin. The PPO agent was trained on a simulated MDP whose transition dynamics are a rough approximation of real player behaviour. The Hexad profiling is rule-based and heuristic, not empirically validated against questionnaire ground truth. The LLM (Phi-3.5 Mini, 3.8B parameters) is small enough to run locally but constrained enough to limit narrative quality. The study design captures a single 30-minute session, missing any cumulative effects. The testbed game is a simplified prototype, not a commercial title. And the sample ($N = 50$) is adequate for medium effects but blind to small ones. These are genuine constraints, not polite scope qualifications, and they define the most productive directions for the work that follows.

6.3 Future Work

Several directions follow naturally from the limitations above, and each would strengthen or extend the system in a distinct way.

Empirical validation of telemetry-derived Hexad profiling. The most immediate and arguably most important extension is a validation study that administers the standard Hexad questionnaire alongside the telemetry-derived profiles for the same participants and quantifies the correlation between the two. This would establish (or refute) the construct validity of the behavioural proxy: the claim that what the system infers from telemetry is meaningfully related to what a validated questionnaire would measure.

Online RL adaptation. The current PPO agent is trained entirely in simulation and frozen at deployment time. An online or hybrid approach, in which the agent continues to update its policy based on real player interaction data collected during live sessions, would produce a policy increasingly tailored to the actual game environment and the actual player population. The challenge is managing the exploration–exploitation trade-off: the agent needs to explore suboptimal actions occasionally to improve its policy, but each suboptimal action is experienced by a real player and may degrade their experience. Safe RL techniques or a warm-start-then-fine-tune approach would be natural candidates.

Larger LLMs and fine-tuning. Replacing Phi-3.5 Mini with a larger open-weight model (e.g., Llama 3.1 8B or a domain-fine-tuned narrative generation model) could substantially improve generation quality, character voice consistency, and emotional range. Fine-tuning on a corpus of high-quality game dialogue would further anchor the model's output in the conventions of the medium.

Multi-session study. A longitudinal study spanning three to five sessions per participant would reveal whether narrative personalization produces cumulative benefits; whether, for instance, the Hexad profile becomes more accurate over time and the narrative increasingly resonates with the player's actual preferences.

Evaluation in a commercial game context. Collaborating with an independent game developer to deploy the plugin in an in-development commercial title would provide a more ecologically valid test, addressing the testbed limitation head-on.

Voice and audio integration. Integrating text-to-speech generation with character-specific voice parameters would move the system from text-only narrative injection toward a richer multimodal experience, raising new challenges around voice consistency and latency.

Explainability and player agency. A transparency mode in which the player can see (optionally) why a particular narrative action was selected ("the system noticed you've been exploring and offered you a lore reward") could be studied as a mechanism for building player trust and a potential design pattern for AI-augmented game narrative more broadly.

6.4 Reflective Report

What follows is a candid account of how the project actually went: what worked, what did not, and what I would do differently.

6.4.1 Project Management and Planning

The project ran for roughly seven months, from initial reading in September 2025 to submission in April 2026. The decision to use a locally hosted LLM (Ollama with Phi-3.5 Mini) rather than a cloud API was made early and proved to be one of the most consequential choices: it eliminated network dependency and data governance concerns but introduced generation quality constraints that shaped the prompt engineering effort for the rest of the project.

Time management was a persistent problem. Integrating five modules across three technology stacks (Python, C#, Ollama) took longer than I expected; each interface worked in isolation but broke in combination.

Figure 6.1 shows the project timeline as it actually unfolded. Major phases overlap deliberately: implementation began before design was finalised, and thesis writing ran alongside development. Milestones (red diamonds) mark proposal approval, system completion, and final submission.

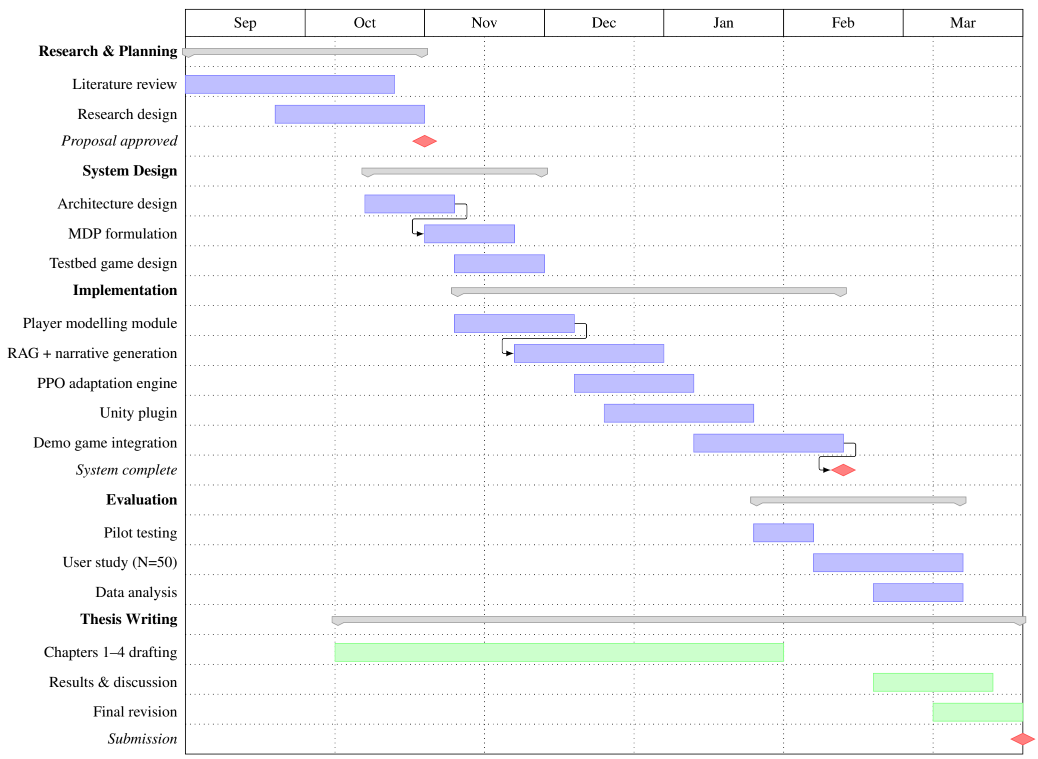


Figure 6.1: Project Gantt chart (September 2025 – March 2026).

6.4.2 Technical Challenges and Learning

Three technical problems stand out. The original plan used ChromaDB as the vector database for RAG, but ChromaDB's Pydantic v1 dependency did not work with Python 3.14. I pivoted to in-memory numpy-based cosine similarity, which turned out to be more than adequate for the small lore corpus but cost me a week of debugging before I accepted that the dependency was the problem, not my code. Lesson: check compatibility before building on a library.

The reward function for the PPO agent went through several failed versions. Without the repetition penalty, the agent converged on a degenerate policy: it selected the same action every step regardless of state. Adding the penalty and the shaped rewards (streak bonus, transition bonus) fixed training stability, but arriving at those additions required reading more RL reward-shaping literature than I had planned for.

Getting the LLM to produce valid JSON consistently was harder than I expected. Phi-3.5 Mini sometimes emits trailing commas, missing quotes, or narrative text outside the JSON structure. The final solution (JSON format mode, explicit schema in the prompt, and a parsing-with-fallback pipeline) was the result of trial and error rather than principled design.

6.4.3 Research Skills Development

Before this project, my experience with research methodology was mostly theoretical. Designing the evaluation (selecting instruments, specifying hypotheses, planning the statistical tests, writing the interview guide) turned abstract knowledge into practical skill. Learning to apply Reflexive Thematic Analysis (Braun and Clarke, 2022) and to plan for Bonferroni-corrected multiple comparisons were the two methodological areas where I grew most.

The literature review taught me to read for gaps rather than summaries. Maintaining the structured comparison table (Table 2.1) forced me to ask, for each prior work, "what is missing here?" rather than "what did they do?", a shift in reading habit that made the research gap much easier to articulate.

6.4.4 Ethical Awareness

Working with human participants made ethical considerations concrete rather than abstract. Designing the consent process and planning the debrief were straightforward, but one question gave me genuine pause: is AI-driven narrative personalization a form of behavioural manipulation? The system is designed to keep players in a Flow state, to make the game harder to put down. That is the stated goal, but stating it plainly reveals an ethical dimension I had not fully considered at the outset.

6.4.5 Key Takeaways

If I were to do this again, three things would change. I would run a small pilot study earlier; even three or four participants would have surfaced the telemetry threshold issues that I only discovered late. I would write automated integration tests for the Python–Unity WebSocket connection from the start, instead of debugging it manually for hours each time something broke. And I would begin the literature review with a systematic search strategy (PRISMA or similar) rather than snowballing from a handful of seed papers, which left blind spots I only discovered when a reviewer pointed them out.

This has been the hardest project of my undergraduate degree, not because any single module was beyond reach, but because making five modules from different domains (RL, NLP, game design, psychometrics, research methodology) work together required a kind of systems thinking that no individual course had prepared me for. I am more interested in AI and interactive media now than when I started, and better equipped to pursue that interest.

6.5 Closing Remarks

The aim of this work is, at bottom, straightforward: to show that large language models, reinforcement learning, and semantic retrieval can be composed into a system that makes a game's story pay attention to the person playing it. Not in the branching-dialogue sense, where paying attention means offering a choice between Door A and Door B, but in the deeper sense of continuously observing how a player engages with the world and adjusting the narrative to meet them where they are. The closed-loop system described in this dissertation is a proof of concept: a demonstration that the integration is feasible, that the adaptation cycle can run within real-time latency constraints, and that the individual modules can talk to each other reliably enough to sustain a coherent narrative across a 30-minute play session.

Whether the result is something players actually notice and value, whether the story's attention registers as personalization rather than as noise, is an empirical question that the evaluation study has begun to answer. But the harder questions are not empirical at all. What does it mean for a story to "understand" a player? Which behavioural signals are worth attending to, and which cross the line from personalization into surveillance? Is a system designed to sustain Flow, to make the game harder to put down, ethically equivalent to one designed to sustain revenue? These are design and ethics questions, not engineering questions, and this work does not pretend to resolve them. What it does provide is a concrete, working system against which they can be meaningfully asked.

Bibliography

AnyRPG (2024). AnyRPG Core: Open-source RPG engine for Unity. GitHub. Available at: https://github.com/AnyRPG/AnyRPGCore.

Ashby, T., Webb, B. K., Knapp, G., Searle, J., and Fulda, N. (2023). Personalized quest and dialogue generation in role-playing games: A knowledge graph- and language model-based approach. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*, pages 1–20. ACM.

Bartle, R. (1996). Hearts, clubs, diamonds, spades: Players who suit MUDs. *Journal of MUD Research*, 1(1). Available at: https://mud.co.uk/richard/hcds.htm.

Braun, V. and Clarke, V. (2022). *Thematic Analysis: A Practical Guide*. SAGE Publications.

Brockman, G., Cheung, V., Pettersson, L., Schneider, J., Schulman, J., Tang, J., and Zaremba, W. (2016). OpenAI Gym. *arXiv preprint arXiv:1606.01540*.

Brown, T. B., Mann, B., Ryder, N., Subbiah, M., Kaplan, J., Dhariwal, P., Neelakantan, A., Shyam, P., Sastry, G., Askell, A., et al. (2020). Language models are few-shot learners. *Advances in Neural Information Processing Systems (NeurIPS)*, 33:1877–1901.

Chen, J. (2007). Flow in games (and everything else). *Communications of the ACM*, 50(4):31–34.

Climent, L., Longhi, A., Arbelaez, A., and Mancini, M. (2024). A framework for designing Reinforcement Learning agents with Dynamic Difficulty Adjustment in single-player action video games. *Entertainment Computing*, 50:100686.

Cohen, J. (1988). *Statistical Power Analysis for the Behavioral Sciences*. Lawrence Erlbaum Associates, 2nd edition.

Compton, K., Kybartas, B., and Mateas, M. (2015). Tracery: An author-focused generative text tool. In *Interactive Storytelling: 8th International Conference on Interactive Digital Storytelling*, pages 154–161. Springer.

Creswell, J. W. and Plano Clark, V. L. (2018). *Designing and Conducting Mixed Methods Research*. SAGE Publications, 3rd edition.

Csikszentmihalyi, M. (1990). *Flow: The Psychology of Optimal Experience*. Harper & Row.

Devlin, S., Hofmann, K., and Birch, D. (2024). player2vec: A self-supervised approach to player embedding from behavioural sequences. In *Proceedings of the AAAI Conference on Artificial Intelligence and Interactive Digital Entertainment (AIIDE 2024)*. AAAI Press.

Faul, F., Erdfelder, E., Lang, A.-G., and Buchner, A. (2007). G*Power 3: A flexible statistical power analysis program for the social, behavioral, and biomedical sciences. *Behavior Research Methods*, 39(2):175–191.

Fountas, Z., Sylaidi, A., Bhatt, U., and Bhatt, P. (2024). Human-like episodic memory for infinite context LLMs. *arXiv preprint arXiv:2407.09450*. Accepted at ICLR 2025.

Hamari, J., Koivisto, J., and Sarsa, H. (2014). Does gamification work? A literature review of empirical studies on gamification. In *Proceedings of the 47th Hawaii International Conference on System Sciences*, pages 3025–3034. IEEE.

Hunicke, R. and Chapman, V. (2004). AI for dynamic difficulty adjustment in games. In *Proceedings of the AAAI Workshop on Challenges in Game AI*, pages 91–96. AAAI Press.

IJsselsteijn, W. A., de Kort, Y. A. W., and Poels, K. (2013). The Game Experience Questionnaire. Technical report, Technische Universiteit Eindhoven, Eindhoven.

Kreminski, M., Samuel, B., Melcer, E., and Wardrip-Fruin, N. (2019). Evaluating AI-based games through the lens of interactivity. In *Proceedings of the AAAI Conference on Artificial Intelligence and Interactive Digital Entertainment (AIIDE)*, pages 59–65. AAAI Press.

Kumaran, V. and Riedl, M. O. (2025). LIGS: LLM-integrated game storytelling. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*. ACM.

Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., Küttler, H., Lewis, M., Yih, W.-t., Rocktäschel, T., Riedel, S., and Kiela, D. (2020). Retrieval-augmented generation for knowledge-intensive NLP tasks. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 33, pages 9459–9474.

Lopes, R. and Bidarra, R. (2011). Adaptivity challenges in games and simulations: A survey. *IEEE Transactions on Computational Intelligence and AI in Games*, 3(2):85–99.

Mateas, M. and Stern, A. (2003). Façade: An experiment in building a fully-realized interactive drama. In *Proceedings of the Game Developers Conference (GDC)*. CMP Media.

Microsoft (2024). Phi-3.5 Mini Instruct model card. Microsoft Research, Hugging Face. Available at: https://huggingface.co/microsoft/Phi-3.5-mini-instruct.

Mnih, V., Kavukcuoglu, K., Silver, D., Rusu, A. A., Veness, J., Bellemare, M. G., Graves, A., Riedmiller, M., Fidjeland, A. K., Ostrovski, G., et al. (2015). Human-level control through deep reinforcement learning. *Nature*, 518(7540):529–533.

Nakamura, J. and Csikszentmihalyi, M. (2002). The concept of flow. In Snyder, C. R. and Lopez, S. J., editors, *Handbook of Positive Psychology*, pages 89–105. Oxford University Press.

Raffin, A., Hill, A., Gleave, A., Kanervisto, A., Ernestus, M., and Dormann, N. (2021). Stable-Baselines3: Reliable reinforcement learning implementations. *Journal of Machine Learning Research*, 22(268):1–8.

Riedl, M. O. and Bulitko, V. (2013). Interactive narrative: An intelligent systems approach. *AI Magazine*, 34(1):67–77.

Ryan, R. M. and Deci, E. L. (2000). Self-determination theory and the facilitation of intrinsic motivation, social development, and well-being. *American Psychologist*, 55(1):68–78.

Schulman, J., Wolski, F., Dhariwal, P., Radford, A., and Klimov, O. (2017). Proximal policy optimization algorithms. *arXiv preprint arXiv:1707.06347*.

Short, E. (2019). Procedural generation and the problem of authorial intent. In *Proceedings of Foundations of Digital Games (FDG 2019)*. ACM.

Souza, V. F., de Lima, C. H., and Chaimowicz, L. (2025). Leveraging Large Language Models for Player Modeling. In *Anais do Simpósio Brasileiro de Games e Entretenimento Digital (SBGames)*, pages 1–10.

Todd, G., Earle, S., Nasir, M. U., Green, M. C., and Togelius, J. (2024). PANGeA: Procedural artificial narrative using generative AI. In *Proceedings of the AAAI Conference on Artificial Intelligence and Interactive Digital Entertainment (AIIDE 2024)*. AAAI Press.

Tondello, G. F., Wehbe, R. R., Diamond, L., Busch, M., Marczewski, A., and Nacke, L. E. (2016). The gamification user types Hexad scale. In *Proceedings of the Annual Symposium on Computer-Human Interaction in Play (CHI PLAY 2016)*, pages 229–243. ACM.

Värtinen, S., Hämäläinen, P., and Guckelsberger, C. (2024). Generating role-playing game quests with GPT language models. *IEEE Transactions on Games*, 16(1):127–139.

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., and Polosukhin, I. (2017). Attention is all you need. *Advances in Neural Information Processing Systems (NeurIPS)*, 30.

Vornhagen, J. B., Thue, D., and Göbel, S. (2022). miniPXI: Development and validation of an eleven-item measure of player experience. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems (CHI 2022)*. ACM.

Wang, W., Wei, F., Dong, L., Bao, H., Yang, N., and Zhou, M. (2020). MiniLM: Deep self-attention distillation for task-agnostic compression of pre-trained transformers. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 33, pages 5776–5788.

Yannakakis, G. N. and Hallam, J. (2009). Real-time game adaptation for optimizing player satisfaction. *IEEE Transactions on Computational Intelligence and AI in Games*, 1(2):121–133.

Yee, N. (2006). Motivations for play in online games. *CyberPsychology & Behavior*, 9(6):772–775.

Zhu, J., Ontanon, S., and Earle, S. (2023). Player modeling for interactive narrative: A survey. *IEEE Transactions on Games*, 15(4):514–528.

Appendix A

CODE REPOSITORY

The complete source code for this project is available at the following repository:

`https://github.com/Rivi9/dynamic-quest-gen`

The repository contains:

- `backend/` — FastAPI Python backend (player modelling, narrative generation, RL adaptation)
- `unity-plugin/` — UPM package (telemetry collection, narrative management, content injection)
- `demo-game/` — Unity 6 testbed project
- `lore/` — Game world lore documents (RAG corpus)
- `evaluation/` — Questionnaires, interview guide, and statistical analysis scripts
- `data/` — Trained PPO model weights and SQLite database

Note: The full source code is not reproduced in this thesis per submission guidelines. Please refer to the repository for implementation details.

Appendix B

GEQ CORE MODULE

The Game Experience Questionnaire Core Module as administered in this study. Reproduced from IJsselsteijn et al. (2013) for research purposes.

Scale: 0 = Not at all, 1 = Slightly, 2 = Moderately, 3 = Fairly, 4 = Very much

Please answer each question based on your experience during the game session you just played.

#	Item	Scale
1	I felt content	0–4
2	I felt skillful	0–4
3	I was absorbed in the game	0–4
4	I felt happy	0–4
5	This game gave me a bad mood	0–4 (R)
6	I thought it was fun	0–4
7	I was fully occupied with the game	0–4
8	I felt annoyed	0–4 (R)
9	I found the game satisfying	0–4
10	I felt frustrated	0–4 (R)
11	It felt like a whole new world	0–4
12	I lost track of time	0–4
13	I felt bored	0–4 (R)
14	I felt good	0–4
15	I was good at this game	0–4
16	I felt tired	0–4 (R)
17	I felt imaginative	0–4
18	I felt that I could explore things	0–4
19	I felt like I had to do things quickly	0–4
20	I had fun with others	0–4
21	I felt challenged	0–4
22	I felt that time went by quickly	0–4
23	I felt pressured	0–4 (R)
24	I felt competent	0–4
25	I felt engrossed	0–4
26	I felt bothered by aspects of the game	0–4 (R)

#	Item	Scale
27	I was deeply concentrated in the game	0–4
28	I forgot everything around me	0–4
29	I felt pleased	0–4
30	I felt completely focused on the game	0–4
31	I enjoyed the game	0–4
32	I felt imaginative	0–4
33	I felt adventurous	0–4

(R) = reverse-scored

Subscale scoring:

Subscale	Items	Notes
Flow (Primary DV, H1)	5(R), 13(R), 25, 28, 31	Mean of 5 items
Immersion (Secondary DV, H2)	3, 12, 18, 19, 27, 30	Mean of 6 items
Positive Affect	1, 4, 8(R), 20, 29	Mean of 5 items
Negative Affect	10, 16, 23, 26	Mean of 4 items
Tension	7, 22, 25	Mean of 3 items
Competence	2, 15, 17, 21, 24	Mean of 5 items

Appendix C

MINIPXI QUESTIONNAIRE

Citation: Vornhagen et al. (2022)

Scale: 1 = Strongly Disagree . . . 7 = Strongly Agree

#	Item
1	I felt in control of my character/avatar
2	I felt a sense of mastery when completing challenges
3	I found the game world interesting to explore
4	I was curious about what would happen next in the game
5	I felt like the game narrative pulled me in
6	I had a sense of presence in the game world
7	I felt free to make meaningful choices
8	The game felt meaningful to me
9	I had positive emotions while playing
10	I felt excited during the game
11	Overall, I enjoyed playing this game

Subscales: Mastery (1, 2); Curiosity (3, 4); Immersion (5, 6); Autonomy (7, 8); Positive Affect (9, 10, 11). Total score: mean of all 11 items.

Appendix D

NARRATIVE PERSONALIZATION SCALE (NPS-3)

The NPS-3 is a custom three-item scale developed for this study to measure perceived narrative personalization. Cronbach's alpha is reported in Chapter 4 ($\alpha = .76$).

Scale: 1 = Strongly Disagree … 7 = Strongly Agree

#	Item
1	The story felt tailored to how I was playing.
2	The narrative responded to my choices and actions.
3	The game seemed to understand what kind of player I am.

Scoring: Mean of all 3 items. If Cronbach's $\alpha < 0.70$, items are analysed individually.

Appendix E

INTERVIEW GUIDE

Study: AI-Driven Dynamic Narrative Personalization in Games
Duration: ∼20 minutes
Analysis method: Reflexive Thematic Analysis (Braun and Clarke, 2022)

Before the interview: Obtain signed consent; remind participant there are no right or wrong answers; start audio recording.

Section 1 — General Experience (5 min)

1. Can you walk me through what you remember most about the game experience?
2. How would you describe the story and the way it unfolded for you?
3. Were there any moments that stood out to you, positively or negatively?

Section 2 — Narrative and Personalization (8 min)

4. Did the narrative feel connected to the way you were playing? Can you give an example?
5. Were there moments where the story surprised you — or felt unexpectedly relevant?
6. Did any NPCs or dialogue feel particularly fitting or particularly out of place?
7. *(Experimental only)* Did you notice the story adapting to you? How did that feel?
8. *(Control only)* How well did the story match your play style?

Section 3 — Flow and Engagement (5 min)

9. Were there moments where you felt in "the zone" — neither too stressed nor too bored?
10. Were there moments where the game felt too easy, too hard, or just right?
11. How did the story contribute to those moments of engagement?

Section 4 — Closing (2 min)

12. Is there anything about the game or narrative you would like to add?
13. Any suggestions for improving the experience?

Probes: "Can you say more about that?" / "What do you mean when you say [X]?" / "When did that happen in the game?" / "How did that make you feel?"

Appendix F

API SPECIFICATION

The FastAPI backend exposes the following REST and WebSocket endpoints:

Method	**Path**	**Description**
POST	/api/telemetry	Submit telemetry window; persists batch to SQLite
POST	/api/narrative/generate	Generate narrative content for a session
GET	/api/player-model/{session_id}	Get current PlayerModel (Hexad + Flow)
GET	/health	Health check
WebSocket	/ws/telemetry	Low-latency telemetry stream

NarrativeContent response example:


```
{
  "action_taken": "PROVIDE_GUIDANCE",
  "content_type": "dialogue",
  "content": "The path below is treacherous. If you
    still carry the torch, keep it lit.",
  "speaker": "Commander Varis",
  "emotional_tone": "tense",
  "lore_refs": [],
  "fallback": false
}
```


All endpoints return standard HTTP status codes. 422 for malformed requests (Pydantic validation). 503 if Ollama is unreachable (returns fallback NarrativeContent with `"fallback": true`).

Appendix G

MDP FORMAL SPECIFICATION

Formal MDP tuple: (S, A, T, R, γ)

State space S: Box(7,) $\in [0,1]^7$

Dim	Symbol	Range	Description
s_0	flow_score	{0.1, 0.2, 0.3, 1.0}	Encoded Flow state
s_1	csr_norm	[0, 1]	challenge_skill_ratio / 3.0
s_2	h_{explorer}	[0, 1]	Hexad Explorer score
s_3	$h_{\text{socializer}}$	[0, 1]	Hexad Socializer score
s_4	h_{achiever}	[0, 1]	Hexad Achiever score
s_5	$h_{\text{disruptor}}$	[0, 1]	Hexad Disruptor score
s_6	t_{progress}	[0, 1]	session_elapsed / 1800

Reward function $R(s, a, s')$:

$$R = R_{\text{state}}(s') + R_{\text{step}} + R_{\text{streak}}(s', n) + R_{\text{transition}}(s, s') + R_{\text{repetition}}(a, a_{[-3:]})$$

Where:

- R_{state}: FLOW $= +1.0$, BOREDOM $= -0.3$, ANXIETY $= -0.5$, APATHY $= -0.8$
- $R_{\text{step}} = -0.05$
- $R_{\text{streak}} = +0.20$ if $s' =$ FLOW and $n > 2$
- $R_{\text{transition}} = +0.30$ if $s \in \{\text{ANXIETY}, \text{BOREDOM}\}$ and $s' =$ FLOW
- $R_{\text{repetition}} = -0.15$ if all last 3 actions identical

Discount factor γ: 0.95

Episode length: 360 steps (30 minutes at 5-second intervals)

Training: PPO (Stable-Baselines3 v2.x), 200,000 timesteps, 4 parallel environments.